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Information structure from a comparative diachronic perspective

Abstract: This chapter examines the interaction between syntax, morphology and information structure in Germanic from a comparative diachronic perspective. Making use of syntactically annotated corpus data, we examine historical stages of English, Icelandic, High German and Low German and present a quantificational comparison of (i) fronted constituents in the prefield and (ii) object ordering in the postfinite domain, which allows us to draw conclusions with respect to the mapping between word order and information structure for the various languages. We focus in particular on the relation between constituent fronting and topicality, suggesting a view of topic which incorporates the Strawson/Reinhart's notion in the situation-semantic approach. Conceiving of topic as a triple of situation coordinates <individual, time, location> used to assess the sentence, we propose that prefinite pronouns (personal, temporal, locative) indicate ongoing topic situation coordinates, while non-pronominal prefinite locative and temporal adverbials and PPs act as respective coordinates' shifters. We argue that Germanic languages have diachronically drifted, to various extents, from a "local" anchoring strategy of IS characterized by ongoing coordinate marking to a "global" anchoring type, where only shifted coordinates are overtly signalled.

1 Introduction

The role of information structure (IS) in grammatical change in the Germanic languages has drawn a great deal of attention in recent decades (see e.g. the collections edited by Hinterhölzl & Petrova 2009; Meurman-Solin et al. 2012; Bech & Eide 2014a, as well as the overview chapter by Eckardt & Speyer 2016). The booming interest in this area has coincided with the rise of quantitative studies of syntactic change facilitated by syntactically annotated historical corpora for various languages (see Haug 2015; Pintzuk et al. 2017; Eckhoff et al. 2018; Taylor 2020). Accordingly, there have been many corpus-based studies on the diachronic interaction between syntax and IS which have revealed nuanced, language-specific

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insights in this area.¹ However, the broader insights to be gained from a comparative, cross-Germanic approach remain largely unexplored (though see Struik 2022a: chapter 7). This chapter contributes towards filling this gap by looking at parallel diachronic phenomena across several Germanic varieties.

Traditionally it has been assumed that the Germanic languages “lose” discourse configurationality over time. That is, there is a shift from systems where the position of constituents is largely determined by their IS role (discourse configurationality, Kiss 1995; Surányi 2016) to systems where IS plays a less important role in syntax (see e.g. Faarlund 1990: Chapter 4 on Norwegian; Burridge 1993: Chapter 6 on Dutch; Fuß 2008: 20 on English and German; Trips & Fuß 2009 on English).

The view that the Germanic languages have largely abandoned discourse configurationality rests on two series of empirical observations. First, it is well documented that earlier stages of Germanic languages exhibit certain syntactic configurations which are not attested in today’s varieties. For instance, Old and Middle stages of Dutch and English feature IO-across-DO scrambling, unattested in the present-day varieties (e.g. Weerman 1997: 432–433). The second series of observations establishes an association between the “disappeared” configurations and specific IS categories, such as topicality or givenness. For instance, object-verb (OV) orders attested in historical stages of Icelandic and English which are no longer grammatical have been shown to be associated with given objects (e.g. Hróarsdóttir 2008; Struik & van Kemenade 2020) (on the definitions of givenness see §3 below).

Based on these two series of observations, a conjecture is made that the loss of certain syntactic configurations equates to the loss of discourse configurationality. This conjecture, however, has to be nuanced in at least two important respects. On the one hand, it has been established that some of the lost configurations, such as “heavy NP shift”, were not necessarily governed by IS considerations, but by other factors such as prosodic well-formedness (e.g. Taylor & Pintzuk 2015 for Old English; Walkden 2014 for Old Saxon). On the other hand, today’s Germanic varieties still show syntactic alternations which are primarily governed by IS considerations.

It is also widely agreed that different Germanic varieties showcase this development in different ways and to different degrees. In this paper, we conduct a quantificational comparison of the mapping between syntax and information structure across English, High German, Low German and Icelandic, which together exemplify Insular West Germanic, Continental West Germanic and (Insular) North

¹ On English, see e.g. Taylor & Pintzuk (2012); Komen et al. (2014); Struik & van Kemenade (2020); Struik (2022b); Los et al. (2023).

On Icelandic, see e.g. Booth & Beck (2021); Booth (2021, 2024).

On High German, see e.g. Light (2011); Coniglio et al. (2017); Catasso (2021).

On Low German, see e.g. Walkden (2014); Struik (2022b).

Germanic. We structure the comparison along two syntactic parameters. The first of these is the occurrence of constituents in the prefield. Examining the blue morphological and lexical profiles of the most frequent prefield constituents and their pragmatic interpretation, we draw conclusions about their role in relating the proposition expressed by the sentence in question to the background information, that is, their IS role. The second syntactic parameter concerns order permutations of postfinite arguments relative to each other (“scrambling”). We focus on this dimension since it has been previously identified as a locus of change in Germanic. Since this volume focuses on grammatical change, we favour changing aspects of the realisation of IS over stable properties.

The paper proceeds as follows: In §2 we discuss specific issues which arise in the diachronic and comparative analysis of IS phenomena, and in §3 we introduce the IS notions we assume. In §4 we present our methodology, and in §5 we present out results for the prefield, discussing these in the context of previous findings from the literature. In §6 we discuss the postfinite domain, focussing on: (i) IS motivations for object order in ditransitives and (ii) the function of anaphoric adverbs as discourse partitioners. §7 concludes the paper.

2 Challenges for comparative diachronic work on IS

2.1 Diachronic challenges

Labov (1994: 11) famously described historical linguistics as “the art of making the best use of bad data”. The fact that the only data available for earlier diachronic stages is a finite written attestation poses a serious challenge for work on IS, especially compared to synchronic work where acceptability judgements and prosodic cues are at hand to offer insights not available to the historical linguist (see e.g. discussion in Petrova & Solf 2009).

Another major barrier is the fact that, unlike work on contemporary varieties, for past language stages the researcher does not have immediate or full access to the contextual knowledge. As a result, it is usually unclear what could be considered shared knowledge among speakers in a language community in a particular period. This makes it challenging to define the relevant Common Ground, a concept key to IS, as discussed in §3. This problem is often compounded by what Merten & Seemann (2018: 823) refer to as a “comparative fallacy”, i.e. when researchers view historical linguistic data through the lens of a modern language. An example of

this problem was highlighted by Wallage (2016, 2017) for the annotation of IS in historical English corpora.

Many of these empirical problems are reflected in the general lack of IS-annotated corpus data for historical Germanic. While the availability of syntactically annotated corpora for various historical Germanic varieties has produced empirically robust and reproducible findings on syntactic change, studies on IS-related phenomena do not generally have the luxury of such ready-made annotations. Indeed, the fact that there are often definitional discrepancies regarding IS concepts (see §3) poses a challenge for the annotation of IS in general; corpus annotation is essentially an identification and classification task, and so one must be able to come up with a sensible set of well-established categories which are easily operationalisable via firm, generally agreed diagnostics. Various challenges have been acknowledged in this context, both for the annotation of contemporary varieties (e.g. Cook & Bildhauer 2011, 2013; Bohnet et al. 2013), as well as specifically for historical data (e.g. Bech & Eide 2014b; Haug et al. 2014; Lüdeling et al. 2016; Taylor 2020; Booth 2022).

As such, researchers interested in comparative diachronic studies of IS in Germanic have three possibilities, which may need to be combined to ensure the most robust empirical outcomes. Firstly, one can make use of the sparse resources which do feature direct IS annotation of historical Germanic texts. The ISWOC treebank (Information Structure and Word Order Change in Germanic and Romance Languages, Bech & Eide 2014c) includes Old English data where referential noun phrases are annotated for information status following the scheme developed by Haug et al. (2014). The Tatian Corpus of Deviating Examples (T-CODEX, Petrova et al. 2009) provides IS annotation of parts of the Old High German Tatian, with information on givenness and aboutness, as well topic-comment and background-focus articulations.

Secondly, one can hand-annotate data from pre-existing corpora which does not come with IS annotations for the particular properties one is interested in. Although this is time-consuming and requires a high level of theoretical understanding of IS and knowledge of the historical language stage, such an approach can yield important insights (e.g. Hróarsdóttir 2008 on historical Icelandic; Taylor & Pintzuk 2014 on Old English; Petrova 2015 on Middle Low German; Coniglio et al. 2017 on Old High German). The program Cesax, developed by Komen (2011, 2012, 2013b), has been specially designed to allow for semi-automatic coreference resolution of syntactically annotated treebank data, and has been used in IS-related diachronic studies of the English (e.g. Komen 2013a; Los et al. 2023).

Thirdly, one can resort to indirect evidence, exploiting more reliable types of annotation other than direct IS annotation, which can be used as proxies to study IS indirectly, e.g. certain syntactic structures and morphological forms, (see discussion

in Lüdeling et al. 2016). Such an approach has been applied to IS-driven syntactic change in, for example, Icelandic (Booth & Beck 2021), where pronominalisation and definite marking are used as proxies for continuing topics.

Another issue to bear in mind is that the historical attestation of the Germanic languages typically involves biases towards particular text types in individual stages, thus hindering fair diachronic comparisons. For instance, the IcePaHC corpus of historical Icelandic (Wallenberg et al. 2011) contains texts from various genres, but these are not evenly distributed across the diachronic dimension. Most of the texts are narratives, with traditional Icelandic sagas dominating in the early stages and modern fiction in the later stages, while religious texts and biographies make up the bulk of the Early Modern texts (see e.g. Schätzle & Booth 2019). Since various IS phenomena are known to interact with genre in a non-trivial way (e.g. Kerz 2012), the genre bias inherent in the diachronic Germanic attestation must be taken into account when analysing data in this area.

2.2 Comparative challenges

A key issue for *crosslinguistic*, *comparative* studies of IS is to what extent IS notions can be assumed to apply universally across languages. This is an important, though often neglected consideration, since many works on IS appear to implicitly assume a crosslinguistically applicable inventory of IS primitives, which vary from language to language only in terms of their formal expression (e.g. prosody, position in the sentence, morphological marking). For instance, many key works on IS assume that topic and focus are basic IS primitives which can be universally defined across languages (e.g. Lambrecht 1994: chapter 4–5; Vallduví & Engdahl 1996; Erteschik-Shir 2007: chapter 2; Krifka 2008; Féry & Ishihara 2016; Song 2017: chapter 3).

However, empirically speaking it remains an open question as to whether there is crosslinguistic variation in the precise IS categories employed *per se*, beyond their formal encoding (see e.g. discussion in McNally 1998; Öhl 2010; Matić & Wedgwood 2013). For instance, McNally (1998) points out that the universality of IS (“information packaging instructions” in her terms) cannot be necessarily assumed, but rather that there may be variation, albeit slight, in the precise nature of the information packaging instructions which specific varieties encode. Specifically for topics, Öhl (2010: 231) notes that, since topics can have a range of different IS properties, topicality should not be viewed as a primitive as such but rather that there are primitive features of “perspectivation” which determine what acts as topic. Similarly, Matić & Wedgwood (2013) argue that, although the crosslinguistic phenomena generally grouped under the term “focus” are comparable in the sense

that they involve similar interpretative effects, they can potentially arise via diverse underlying mechanisms. As such, rather than attempting to identify apparent instances of a pre-established focus category across varieties, Matić & Wedgwood (2013) advocate an approach whereby one examines crosslinguistic data for the similarity of these effects and modes of convergence, without *a priori* relating this to one single mechanism or category.

Since the crosslinguistic applicability of IS categories remains an open question, in this paper, rather than assuming that a given IS notion is relevant for all languages at all stages and looking for ways it is realized (e.g. “How is topic expressed in X?”), we adopt a semasiological approach, seeking to determine the information-structural role a particular (morpho-)syntactic configuration plays in a given variety (e.g. “What is the role(s) of the prefield in X?”). Conceptually, when defining the IS role of a given marker, we operate in terms of the basic informational concept of (sub-)proposition and the way it relates to the CG (as laid out in §3).

3 Concepts and definitions

Broadly construed, IS concerns the relationship between the proposition and the utterance context, understood as the mutually assumed knowledge of the speech act participants. Thus, IS has to do with the identification of individuals and events with respect to the state of the shared knowledge (“Common Ground”, Stalnaker 1978). This can concern various dimensions, including temporal, spatial, personal, evidential and modal aspects. At least two major axes of identification have been discussed in this context: i) whether the information about the existence of a participant/event is new/old (if not new, mentioned/non-mentioned before); ii) whether a participant/event has relevant alternatives or not. As Reinhart (1981: 61) notes, it is conceptually unclear what it means for a participant to be new/old information. We will therefore talk instead about relevant existential propositions, which can be new/old information, rather than about old/new participants or NPs. An expression *old/given NP* will be used only as a shortcut for “NP which is associated with an existential presupposition that an individual with the property denoted by the NP exists at the relevant time and space indices”.

Despite the terminological variation, most current definitions of IS seem to mention the following three key ingredients: i) the propositional content; ii) the speaker’s intentions concerning conveying the propositional content; iii) the (speaker’s conception of the) listener’s mental state with respect to the relevant propositional content.

For instance, Vallduví (1992: 15) defines information packaging as “a small set of instructions with which the hearer is instructed by the speaker to retrieve the information carried by the sentence and enter it into his/her knowledge store”, which expands on Chafe’s (1976: 28) original “ways in which a speaker accommodates his speech to temporary states of the addressee’s mind”. Very similarly, Matic (2015: 95) defines IS as “a subfield of linguistic research dealing with the ways speakers encode instructions to the hearer on how to process the message relative to their temporary mental states” (see Kroeger 2017 for a short overview of essential definitions).

There is a greater degree of conceptual discord with regard to the precise nature and inventory of the “instructions” that the speaker gives to the hearer concerning the propositional content. Using Krifka’s (2008) metaphor of IS as “Common Ground (CG) management”, we can distinguish between two primitive relations between a proposition and the Common Ground that the speaker manages: i) CG entails the proposition in question; ii) CG does not entail the proposition in question. These relations do not, however, appear to be necessarily defined on the basis of fully-fledged propositions denoted by whole sentences (Tonhauser 2012). In fact, these are outlier cases. Sentences denoting propositions that are fully entailed by the CG are only of rhetorical use, and sentences denoting propositions that are fully not entailed by the CG are known as *thetic*, also not the most frequent sentence type (cf. Lambrecht 1994, “sentence focus”).

Instead, it appears that most frequently the speaker indicates the relation between the CG and a number of “sub-propositions”. To take the example in (1) from Kroeger (2017: 3), *that* and *my* can be assumed to signal that the existential (sub-)propositions *there exists a man at given time and space indices* and *there exists a dog owned by the speaker at given time and space indices* are entailed by the CG. Morphemes associated with such existential sub-propositions are known as *presupposition triggers*. Nothing is signaled, either syntactically or morphologically, concerning the existence of an event of kicking with the man in question and the speaker’s dog as its participants, which is arguably a default option for non-entailed (sub-)propositions concerning the main event or predicate focus (Lambrecht 1994).

(1) That man just kicked my dog!

Existential (sub-)propositions often involve deictically defined parameters (can be said to be deictically “anchored”). In (1), a demonstrated spatial domain in the case of *that* and the speaker in the case of *my* are deictic parameters. In other cases, a (sub-)proposition that is signalled to be entailed by the CG can have an anaphorically defined parameter (or anaphoric deictic, if anaphora is considered to a sub-case of deixis, as in Bühler’s 1934 original definition).

- (2) Yesterday I finished work at 16:00. Then I went for a short walk.

We assume that the second sentence in (2) involves as part of its denotation a sub-proposition (triggered by *then* as a presupposition) that there exists a time t which immediately follows the anaphorically given time t' , which corresponds to the time t' at which the proposition denoted by the previous sentence holds. While the existence of a time (interval) which follows the just mentioned interval is trivial, because there is an immediate successor to any time interval, (2) conveys, as new information, that the proposition it denotes holds with respect to the time interval so defined.

One of the most widely used and psychologically real IS notions is that of topic, though it is possibly also the most controversial formally. Already Reinhart (1981: 56) writes that “almost every imaginable approach to its definition has been actually proposed”. Reinhart’s own influential notion of “aboutness topic” is based on Strawson’s idea that “assessment of statements as true or false are commonly...topic centered” (Strawson 1964, cited from Reinhart 1981: 56). The aboutness topic corresponds to the NP whose extension serves as the assessment domain for the proposition denoted by the sentence in question: the truth of the relevant proposition is assessed for all the elements in the extension of the topic NP. The notion of aboutness topic approaches the concept of Austinian topic situation, understood as the situation a given sentence is about (Austin 1950; Schwarz 2019, Kratzer 2023). We bridge the conceptual gap by proposing that elements empirically identified as Reinhart’s aboutness topics serve to establish *the participant coordinate of the topic situation*, the latter being also specified with time and space coordinates.

In addition to aboutness topics, the literature also distinguishes so-called *frame- or scene-setters*, characterised in Speyer (2019: 340) as “expressions that define the situation in which the proposition is evaluated with respect to its truth value”. Similarly to Speyer (2010), Frazier & Clifton Jr. (2018) argue that, in Present-day English, clause-initial temporal and locative PPs set topic situations. Their hypothesis is given in (3) (Frazier & Clifton Jr. 2018: 175).

- (3) **Topic situation hypothesis:**

Initial temporal and locative PPs in English introduce topic situations. By default, following material is included in the topic situation until a new topic situation, or incompatible information, is encountered. Other things equal, the topic situation supplies the preferred context for implicit domain restriction.

Synthesising the notions of “aboutness” topics and temporal and locative topic situation setters, we will approach topics as n-tuples of interpretation parameters

consisting of (a subset of) participant, time, and space coordinates. While the traditional “aboutness topics” set the participant coordinate, “scene-setters” do so with respect to time and space coordinates. In general, we will be assuming two major IS strategies with respect to topic coordinates: indicating an ongoing value of a given coordinate and indicating a coordinate shift.

Modifying the definition of topic in Krifka (2008: 41), we assume (4) as our approach to topic.

(4) **Topic (situation)**

The topic (situation) is a n-tuple of coordinates (individual and/or temporal and/or locative) which determine where in the CG the information expressed in the comment constituent should be stored and with respect to which its truth conditions should be evaluated.

We propose that, in the case of participant and space coordinates, the ongoing value corresponds to the value of the topic coordinate in the previous sentence, or an immediate context value. On this view, so-called “discourse pro-drop”, as in Mandarin, is the strategy of not overtly expressing the default value of the participant coordinate. The time coordinate, by default, takes the value of the time interval that immediately follows the time interval with respect to which the previous sentence is interpreted, because of the natural isomorphism between narrative and temporal sequences. We propose that adverbs which can be translated as “then” indicate the default value of the topic time coordinate. We further propose that preverbal anaphoric temporal adverbs set a topic time coordinate, with respect to which eventualities in the current proposition are evaluated by default. By default, the topic time immediately follows the previously established topic time interval.

A change in participant, space or time coordinate seems to be more commonly, if not universally, signalled. Particularly relevant notions here are *poset* (Hirschberg 1991), i.e. “a set defined by a transitive partial ordering relation such that its members may be related either as higher/lower values within the set or as unordered alternates at the same level” (Birner 2004: 54) and Rooth’s focus as an association between a constituent and a set of alternatives to its denotation (Rooth 1992).

4 Methodology

For the diachronic comparison of Germanic in this paper, we focus on historical data from four languages, for which diachronic Penn-style treebanks are available: English (Insular West Germanic), High German and Low German (Continental

West Germanic), and Icelandic (Insular North Germanic). The relevant corpora and their specific time spans are outlined in Table 1. All corpus queries, datasets and codes employed in the analyses are available on the Open Science Framework.²

Language	Corpora
English	YCOE (800–1169, Taylor et al. 2003) PPCME2 (1170–1500, Kroch et al. 2000) PPCEME (1501–1698, Kroch et al. 2004) PPCMBE (1706–1913, Kroch et al. 2016)
Icelandic	IcePaHC (1150–2008, Wallenberg et al. 2011)
High German	IPCHG (1360–1650, Sapp et al. 2024)
Low German	CHLG (1279–1580, Booth et al. 2020)

Tab. 1: Languages and corpora examined in this paper

As mentioned in §1, in this paper we examine two separate domains in the clause: i) the prefinite domain (prefield) and ii) the postfinite domain. For each of these domains, we extract readily available lexical and morphosyntactic information from the corpora in Table 1 which we believe can be informative as to the interaction between information structure and syntax. For example, we assume that certain adverbials, e.g. temporal anaphoric adverbials (‘then’, ‘at that time’), semantically relate the event time of the proposition in question to the time of some other event whose existence is established in the CG, i.e. establish/maintain the topic situation, see (4). In this way, we leverage the pre-existing annotations in each of the corpora to shed light on the role of information structure in the grammar. As another example, for subjects and objects we use pronominality as an approximate of topicality.

For each language, the desired syntactic information was extracted using coding queries in CorpusSearch (Randall et al. 2004). In particular, we extracted data on our syntactic domains of interest (prefield and postfinite domain). The extracted data were tidied, plotted and visualised using R (R Core Team 2013). When applicable (i.e. if we had a big enough sample), we employed statistical modelling to predict the effect of time over the IS effects on syntax. Otherwise, we report a table with raw frequencies and percentages.

We used logistic models to predict the probability p of a certain construction appearing over time (t).

$$p(t) = \frac{1}{1 + e^{-t}} \quad (1)$$

² <https://osf.io/m3pd5/>

We have deliberately chosen this fitting curve for several reasons. First, logistic models have become the standard for diachronic surveys since Altmann et al. (1983) for capturing emerging patterns which follow an S-shaped curve of linguistic change (Kroch 1989). Conceptually, logistic regression captures the competition of an erstwhile grammatical feature and the emergence of an antagonist competitor in the linguistic system, creating a dynamic process constrained by competition (see Postma 2010 for a more detailed description). Second, even though some diachronic trajectories may be better captured by other distributions (e.g. fitting polynomials) we decided to be consistent throughout our datasets and varieties, employing logistic regressions in order to maintain comparability across varieties and case studies.

Since the statistical analyses were run using the text date as predictor, and since we plotted the probability of a prefield constituent in a text as a single data point, some distinct texts might have become indistinguishable because they were written in the same year, thereby resulting in the same data point. To circumvent this limitation, we corrected the dates by adding one unit to those cases where the date was the same but the text ID was different.

Both model interpretation and probability distributions will be employed to appreciate the changes over time in specific syntactic fields. By analysing these quantitative trends, our goal is to detect whether these dynamic processes are related to changes in the mapping to IS in a specific language and, more broadly, across Germanic.

5 Variation and change in the IS properties of the prefield

In this section, we examine the IS effects of prefield constituent placement. We use the general term prefield to refer to all the position(s) preceding the finite verb in languages which allow for more than one such position. We first compare English (§5.1) and Icelandic (§5.2), two Germanic languages which are generally considered to be particularly innovative and conservative respectively, and for which syntactically annotated corpora are available for very long diachronic spans, allowing us to examine the data over multiple time periods. We then compare the status of the prefield for two closely related varieties, High German (§5.3) and Low German (§5.4), via the historical Penn-style treebanks which are currently available for these varieties, which cover roughly the same time range (High German, 1360–1650; Low German, 1279–1580).

5.1 English

Old and Early Middle English are widely considered to be verb-second (V2). The nature of the first constituent has been mostly characterised in IS terms. In particular, it has been argued that “topic-verb configuration... defines the verb-second phenomenon” (Kroch et al. 2008: 353), but also that the prefinite constituents divide into two classes: “given” and contrastively focused (Los & Dreschler 2012, Speyer 2010). Considering that the rate of null subjects had fallen to around 3% already by Old English (Walkden & Rusten 2017), prefinite placement of a non-subject constituent in a V2 clause implies subject-verb inversion. With the decline of the latter in declaratives during the Middle English period, English left the V2-family (e.g. Haeberli 2007: 17).

The narrowing down of the range of constituent types in the immediately prefinite position to subjects has been related, in particular, to the loss of “local anchoring” (Los & Dreschler 2012; Los & van Kemenade 2018). While local anchoring involves “link[ing] each discourse move to a point in time and space” (Los & Dreschler 2012: 2), under “global” anchoring the scope of the respective coordinates extends over larger stretches of discourse. To make an analogy with a relatively better studied phenomenon of (radical) pro-drop, languages can be divided into those that encode the feature of the topic participant(s) locally, by means of pronouns or verbal endings in every finite clause, and those where they are left locally implicit, assumed to be globally maintained until an explicit topic-shift takes place.

To track the decline of local anchoring in English, Los & Dreschler (2012: 6) compare relative frequencies of prefinite subjects vs. objects and adverbials in historical English, showing that the relative frequency of prefinite subjects grows, at the expense of the other two constituent types. They also demonstrate that the share of PPs containing demonstratives – a proxy for local anchors – among clause-initial PPs declines over time, as does the absolute frequency of clause-initial temporal and locative adverbs meaning “then” and “there” (Los & Dreschler 2012: 8). Below, we build on this approach, keeping track of additional morphosyntactic distinctions among prefinite constituents and distinguishing between $V_{=2}$ and verb-third ($V_{=3}$) clauses.

In addition to the local anchoring hypothesis, prefinite non-subjects in Old English and modern V2 languages have been related to the maintenance of the given-new information flow. In particular, Dreschler (2014) presents quantitative data showing the prevalence of given direct objects among prefinite ones.

In what follows, we unify the “local anchoring” hypothesis concerning prefinite spacial/temporal adverbials and the “given-before-new” account of the prefinite direct objects by subsuming the notions of spatiotemporal anchors and individual-

denoting topics (given information) under the general notion of topic situation coordinates (see (4)).

5.1.1 Prefield: morphosyntactic signatures

English has not been a *strict* V2 language in its recorded history, with a substantial number of finite clauses featuring more than one prefinite constituent across time. As a baseline, consider Figure 1 (left) showing the relative frequencies of main clauses with a finite verb in the first, second, third, fourth or fifth position across time.³

As a first approximation of the changes in the IS role of the prefield, we consider the distribution of prefinite constituent types (see also Los & Dreschler 2012). Specifically, we focus on three (for now, linearly defined) positions: i) initial constituents in $V_{n=2}$ (Fig. 1, right); ii) initial constituents in $V_{n=3}$ (Fig. 2, left); iii) second constituents in $V_{n=3}$ (Fig. 2, right).⁴⁵

3 We excluded clauses with null subjects (elided or otherwise) and expletives (both null and overt) because of subject-driven idiosyncrasies of their IS profile. We use $V_{=n}$ to refer to clauses with the finite verb in the n -th position, and we reserve V2 to denote a type of language where the position before the finite verb in second position is frequently filled by non-subject material. The preverbal negative particle “ne” is ignored when considering the linear position of the finite verb, on the assumption that it is a syntactic head (Kemenade 1997: 152) and hence is not in complementary syntactic distribution with phrasal preverbal constituents we are examining. We also discarded left dislocated constituents as, per definition, being external to the clause.

4 We excluded those types of constituents whose relative frequency in the relevant position* at no time period exceeds 3%.

5 We coded temporal adverbs *þa* (‘then’), *þonne* (‘then’), *eft* (‘afterwards’), *forþ* (‘afterwards’), *sone* (‘soon’) and *nou* (‘now’) as adv-TMP, locative adverbs *þær* (‘there’), *her* (‘here’), *thider* (‘thither’), *hither* (‘here’) as adv-LOC and manner adverbs *þus* (‘thus’), *swa* (‘so’) and *ek* (‘likewise’) as adv-MAN.

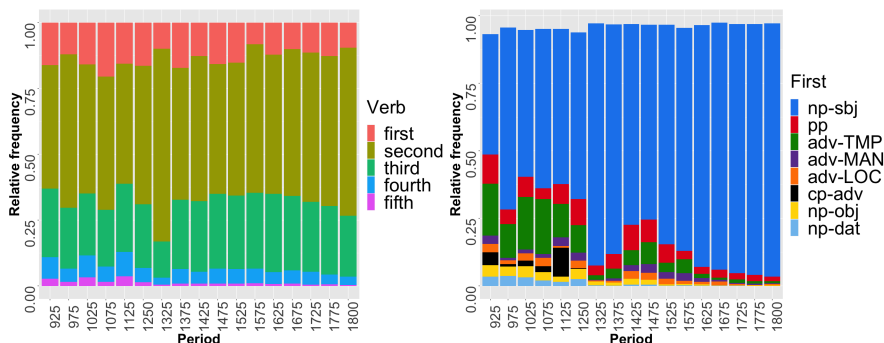


Fig. 1: Finite verb position (left) and initial XPs in main finite $V_{n=2}$ (right) in English

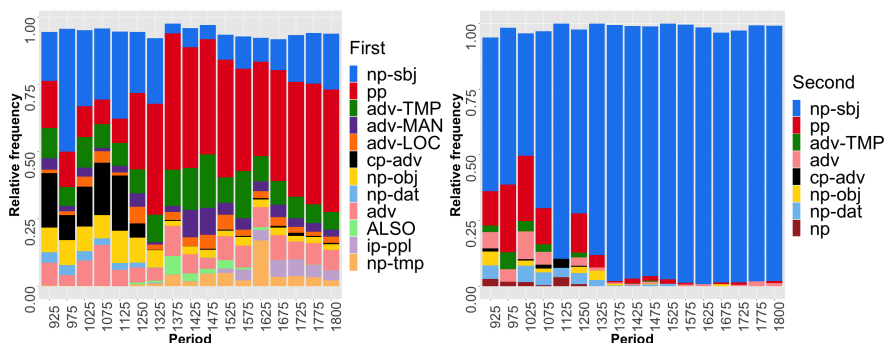


Fig. 2: Initial (left) and second (right) XPs in main finite $V_{n=3}$ in English

In $V_{=2}$ clauses, the most frequent prefield constituents are subjects (increasing in frequency), then adverbials (decreasing), then PPs (decreasing), then direct and indirect objects (decreasing). After 1250, constituents other than subjects become relatively rare in the prefield.

It is in $V_{=3}$ clauses that we can tease apart the effect of the loss of subject inversion and the changes in IS-driven fronting. Here, the most frequent clause-initial constituents are again subjects (decreasing in frequency), then adverbial clauses (disappearing, as adverbial correlatives go out of use in general, Links 2018), then adverbials (stable, pending a formal test), then PPs (increasing), then direct objects (decreasing). The most frequent *immediately* prefinite constituent types in $V_{=3}$ are subjects (increasing), adverbials (decreasing), PPs (decreasing or stable, pending a formal test) and direct and indirect objects (decreasing).

An important interim conclusion is that, while the increasingly frequent placement of subjects in the immediately preverbal position blocks other types of constituents from this position in $V=2$ and $V=3$, especially adverbs, the clause-initial position in $V=3$ has retained its Old English profile to a large extent.

5.1.2 Subjects

We now look at prefinite subjects in isolation, assuming pronominalisation as a proxy for topicality and subjects carrying existential presupposition triggers as a proxy for old information.

To estimate the inversion rate, we fit a logistic regression model to a binary variable *prefinite/postfinite* subject within a subset of finite main $V=2$ and $V=3$ clauses with: i) nominal subjects ($N = 74138$); ii) pronominal subjects ($N = 84808$). The two models are visualised in Fig. 3 (intercept -3.94 , time coefficient 0.0038 , $p < 0.0001$ for nominal and intercept -1.94 , time coefficient 0.0031 , $p < 0.0001$ for pronominal subjects). The probability of being prefinite is higher for pronominal subjects than for nominal ones for the equivalent periods, but the rates of change are extremely similar.

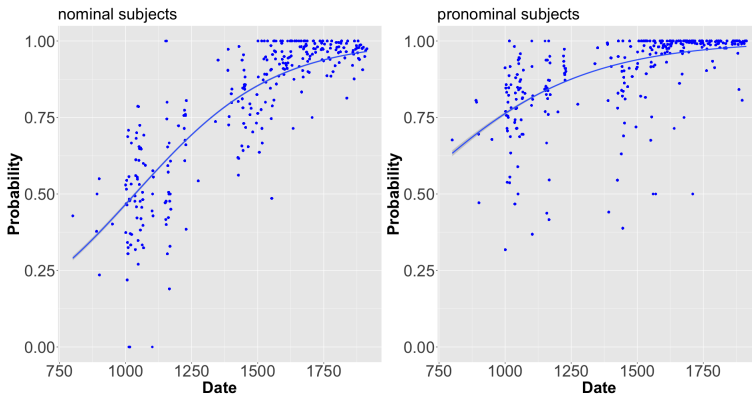


Fig. 3: Logistic regression models of prefinite subject placement in $V=2/V=3$ English

Using pronominalisation as a proxy for topicality (Reinhart 1981), we track subject topicality across time. We fit a logistic regression to a binary variable pronominal/non-pronominal in finite $V=2$ and $V=3$ clauses with either non-pronominal or overt pronominal non-expletive subjects ($N = 174631$). The regression model is visualised in Fig. 4 (intercept -1.129 , time coefficient 0.0008 , $p < 0.0001$).

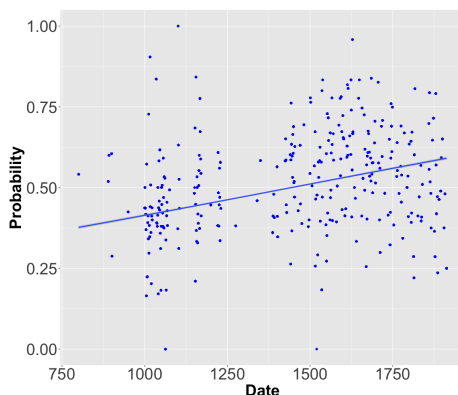


Fig. 4: Logistic regression model of subject pronominalization in $V=2/V=3$ English

The model detects a significant increase in subject pronominalisation. This result is in line with the general hypothesis of Komen et al. (2014) that subjects become more discourse-linking to “compensate” for the decline in other local linkers in the prefield, such as adverbs and topical direct objects. This, however, contrasts with the results of Los & Dreschler (2012), who argue, on the basis of samples from two Bible translations, that the pronominalisation of subjects is less frequent in PDE compared to OE.

5.1.3 Objects

Speyer (2010) concludes that, despite the overall quantitative decrease in preverbal objects, qualitatively there is only one type of object which can no longer be prefinite namely “e-topics”, i.e. pronominal objects. Speyer analyses other instances of prefinite object placement as contrastive topicalisation, which he argues survives until Present-day English.

Stevens & Light (2013) argue that demonstrative pronouns are semantically contrastive and therefore pattern with other contrastive elements, rather than with personal pronouns, which they assume to undergo Speyer’s e-topic fronting. This, they argue, explains why fronting of demonstrative pronouns persists until Present-day English.

Quantitatively, we confirm the conclusion of Speyer (2010) and Stevens & Light (2013) that prefinite placement of pronominal vs. nominal and demonstrative direct objects follows different diachronic trajectories. In Fig. 5, we visualise logistic regression models fit to the binary variable *prefinite/postfinite* in the subsets of: i)

personal pronominal, ii) nominal, and iii) demonstrative pronominal direct objects in main clauses with overt subjects ($N = 15262$, $N = 53675$, $N = 2640$, respectively). The models' parameter estimates are as follows: i) intercept 7.763, time coefficient -0.0071 , $p < 0.0001$, ii) intercept 3.172, time coefficient -0.0042 , $p < 0.0001$, iii) i) intercept 1.896, time coefficient -0.0014 , $p < 0.0001$.

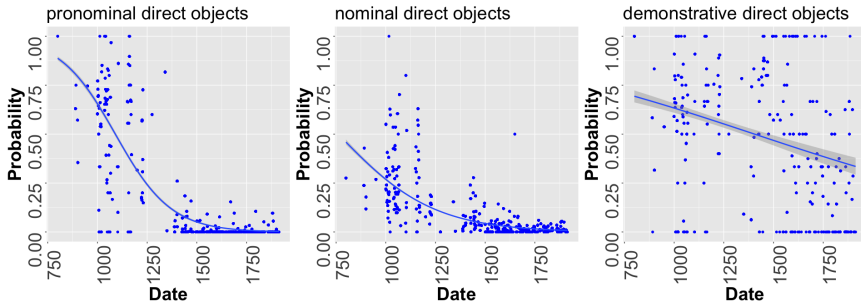


Fig. 5: Logistic regression models of the probability of prefinite nominal, pronominal and demonstrative DO in main clauses with overt non-expletive subjects in English

Based on these models, the probability of fronting declines for different types of direct objects at different rates, which confirms the relevance of the distinction highlighted in Speyer's work. However, in contrast to the claim that topical ($\approx$ pronominal) and contrastive ($\approx$ nominal/demonstrative) object fronting are entirely different phenomena, we unify them under the topic situation hypothesis, distinguishing between current topic coordinates (the former) and topic coordinate shifters (the latter), see §3.

As Fig. 5 (left) shows, pronominal objects are overwhelmingly prefinite in Old English, e.g. (5). This default pattern changes very quickly between Old and Middle English. We assume that the default configuration could not have been associated with a contrastive interpretation.⁶

⁶ It also could not have been produced by in-situ pronominal object placement within a head-final OV: among main clauses where V and T positions could be distinguished, that is, in clauses with an auxiliary (or modal) and participle (or infinitive), there are only 55 object-participle-auxiliary strings vs. 4522 object-auxiliary-participle strings, meaning that prefinite pronominal objects should have moved over T. These data also mean that prefinite object placement cannot be analyzed as scrambling in a T-final clause, pace Stevens & Light (2013: 205).

- (5) & **hit** Englisce men swyðe amyrdon.
 and it English men fiercely prevent.PRT.3PL
 ‘and the Englishmen prevented it fiercely’ Old English,
 Old English, 1153, *cochronE,ChronE_[Plummer]:1073.2.2681* (Speyer 2010: 38)

A series of analyses originating from Frey (2006a,b) (e.g. Speyer 2010; Light 2012; Stevens & Light 2013) analyses the prefinite placement of pronominal objects in historical English as “formal movement”. We depart from this tradition and propose to see prefinite pronominal objects on a par with prefinite pronominal subjects and prefinite pronominal adverbials as overt markers of topic situation coordinates.

We analyze prefinite nominal and demonstrative direct objects as *shifters* of a topic (participant) coordinate. Los & van Kemenade (2018) argue that demonstrative pronouns appear in the Old English prefield in their capacity as topic-shifters (see also Epstein 2011; Breban 2012). A clearly discernible property of clause-initial nominal objects is that their denotation is associated with a salient alternative set. Sequences in (6)–(7) are typical examples of this pattern.

- (6) **Kyng-is** he ouyrcam with uictorye; **psalm-es** he sang
 king-PL he overcome.PRT with victory psalm-PL he sing.PRT
 with melodie; **best-is** he kill-id
 with melody beast-PL he kill-PRT
 “He defeated kings with victory; he sang psalmes with melody; he killed beasts.” Middle English, 1464, (CMCAPCHR,32.8–10)
- (7) **God** wold-est þou haue, and **synne** wold-est þou lacke.
 God will.PRT-2SG you have and sin will.PRT-2SG you lack
 “You would have God and would lack sin.” Middle English, 1425, (CMCLOUD,81.409–410)

5.1.4 Adverbials

As Figs. 1 and 2 show, adverbials are the second most frequent constituent type in the Old English prefield, after subjects. The “adv-TMP” group can be categorised as deictic (including anaphora; see Los & van Kemenade (2018), “discourse-anchoring”). The relative frequency of this group changes only in those positions where they “compete” with the subject, that is, in the immediately preverbal position in $V_{n=2}$ and $V_{n=3}$ clauses; in the initial position in $V_{n=3}$ clauses, the relative frequency of anaphoric adverbs remains roughly the same over time.

It has been observed for English that the semantic contribution of a number of temporal and spatial adverbs, including “then”, depends on whether they are

prefinite or not (Schiffrin 1992; Thompson 1999; Trips & Fuß 2009). For Modern English, this is clear from pairs such as in (8). In (8-a), *then* introduces a time that follows the time of the previous proposition event (accusing), while *then* in (8-b) picks up the accusation time itself.

- (8) Michael accused me of lying yesterday...
- a. ...Then he asked for forgiveness./He then asked for forgiveness.
 - b. ...He looked really agitated then.

The “then”-adverbs in Old English have likewise been associated with a “sequential” interpretation in the prefinite position, e.g. (9-a), and with a co-temporal interpretation in the postfinite domain, e.g. (9-b) (Foster 1975; Trips & Fuß 2009).

- (9) a. *Ponne ærna-ð hy ealle toward þæm feo*
 then run-PRS.3PL they all towards DEF.DAT.SG treasure
 ‘Then they all run towards the treasure.’
 Old English, ca. 900, Orosius:1.17.21.233 (Trips & Fuß 2009: 174)
- b. *gif he hit him ðonne sellan mæge*
 if he it him then give may.PRS.3SG
 ‘if he can give it him then’
 Old English, ca. 890, *cocura*, CP:44.323.24 (Van Kemenade & Los 2009: 235)

On our model, when preverbal, a temporal adverb denotes the time (interval) coordinate of the current topic situation, which by default follows the time interval of the most recently introduced proposition, hence sequential reading. When postverbal, it is interpreted as “simply” a free time (interval) variable, which gets its value in a way other free pronouns do, from the closest available antecedent, which often (but not necessarily) happens to be a time expression in the previous clause (cf. Thompson 1999: 158).

We thus analyse prefinite temporal anaphoric adverbials as establishing the topic time coordinate or, equivalently, the time of the topic situation. As already mentioned, their relative frequency in the prefield appears stable in the positions where they are not ousted by the subject. The same can be said about non-temporal adverbials (*adv-LOC* and *adv* in Figs. 1 and 2). In addition to their relative frequency in the prefield, we check how frequently they appear prefinately as opposed to postfinately. Fitting a logistic regression to the variable *preverbal/postverbal* (temporal adverb) within a subset of finite main $V=2$ and $V=3$ with overt non-expletive subjects ($N = 19815$), we obtain the model illustrated in Fig. 6 (intercept 0.2069, time coefficient -0.0005 , $p < 0.0001$).

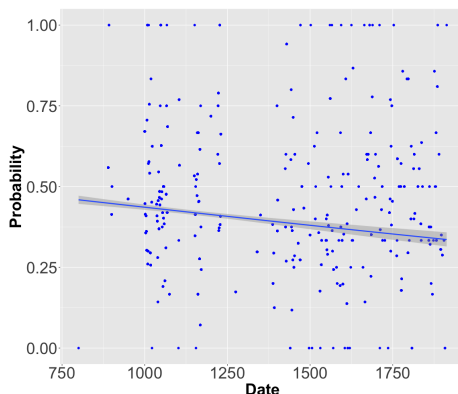


Fig. 6: Logistic regression model of prefinite placement of temporal adverbials in $V=2/V=3$ in English

The model detects a very modest change, in comparison to the changes in subject inversion and the prefinite placement of objects we have seen above. A model fitted to $V=2$ clauses only, the most frequent clause type, does not detect any diachronic change at all (Date coefficient = 0.00005, $p = 0.87$).

We conclude that, with respect to adverbials, the IS changes only to the extent that independent syntactic changes take place. While the relative frequency of adverbials in the prefield is negatively affected by a significant increase in the prefinite subject placement, the estimated probability of a temporal adverbial landing in a prefinite, as opposed to a postfinite position, changes relatively little in $V=3$ and not at all in $V=2$.⁷

5.1.5 PPs

Speyer (2010) considers prefinite PPs, such as *In the afternoon, I usually go for a walk*, as scene-setters. According to Jacobs (2001), the line between those and the aboutness topics is fuzzy, and Frazier & Clifton Jr. (2018) treat prefinite PPs as topic situation setters, while Krifka (2008: 45-46) points out that in the

⁷ Modulo the already mentioned disappearance of adverbial correlatives (Links 2018: 60):

- (i) þa he þa in þæt ealond cwom, þa getimbred he þær mynster
 then he then in that island come then built he there monastery
 ‘On coming to that island, he erected a monastery there.’ Old English, 1037,
 Bede_4:4.272.28.2779

question-answer pairs such as (10), “the topic is Daimler-Chrysler, not Germany or America”, and proposes that frame-setters “choose one out of a set of frames and state that that the proposition holds within this frame” and that they are always focused since “[i]f there is no alternative perspective ... then there is no need for an explicit frame setter”.

- (10) a. How is business going for Daimler-Chrysler?
 b. In [GERmany]_{Frame}, the prospects are [GOOD]_{Focus}, but [in AMERica]_{Frame}, they are [losing MONEY]_{Focus}.

We merge the two views by proposing that locative and temporal non-pronominal PP’s shift topic situation location and temporal coordinates, respectively, in the sense of the definition in (4).

We quantitatively evaluate the diachronic profile of PPs in English in terms of i) the morphosyntactic types of P-complements, and ii) the PPs’ diachronic propensity for prefinite placement. The first question is addressed by means of a simple barplot and the second one via a logistic regression model. Fig. 7 shows the morphosyntactic profile of a total of 17,652 prefinite PPs in finite V₌₂ or V₌₃ clauses with an overt non-expletive subject.⁸

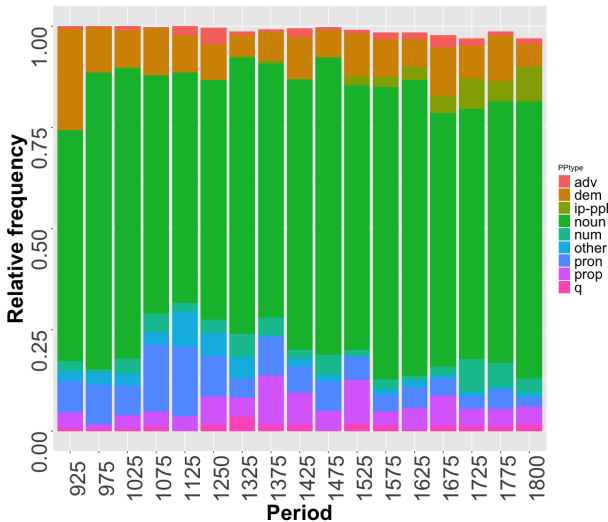


Fig. 7: Relative frequency of P-complement types in prefinite PPs in finite V₌₂ or V₌₃ with an overt non-expletive subject in English

⁸ Again, we discard the types whose relative frequency at no time period exceeds 3%.

The diachronic morphosyntactic typology shows a decline in the relative frequency of pronominal P-complements, conceivably due to the emerging restriction on prefinite (indirect) object placement, which used to be possible in Old English, e.g. (11)–(12).

- (11) & heo for hine gebæd-on:
and PL.NOM for 3SG.MASC.ACC pray-PRET.3PL
‘and they prayed for him’ (cobede,Bede_3:9.186.16.1875)
- (12) & him to geheht Cedd þone halga-n
and 3sg.dat to call.pret.3sg Cedd def.3SG.MASC.ACC holy-SG.OBL
God-es mon
God-sg.gen man.SC.NOM/ACC
‘and (he) called Cedd, God’s holy man’ (cobede,Bede_3:16.226.8.2312)

This result is line with the finding of Los et al. (2023) about the diachronic decline in the relative frequency of prefinite PP’s whose denotation is related to that of an antecedent by an *identity* relation, which they see as part of the overall decline of local anchoring.

We also fit a logistic regression to a dataset of PPs in finite $V=2$ and $V=3$ clauses with overt non-expletive subjects ($N = 95196$), see Fig. 8 (intercept -0.8631 , time coefficient -0.0005 , $p < 0.0001$).

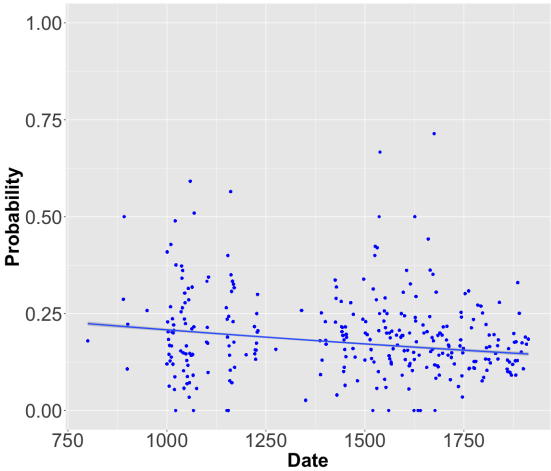


Fig. 8: Logistic regression model of the probability of PP prefinite placement in main clauses with overt non-expletive subjects in English

The model detects a small decrease in prefinite PP placement over time, the coefficient being equal (up to the fourth symbol after the dot) to that of the decrease

in prefinite temporal adverbs. Since indirect object fronting likely accounts for a relatively small share of prefinite PPs, the decline is likely mostly due to the increase in subject fronting, as Fig. 1 (right-hand) illustrates in broad strokes.

5.1.6 Interim summary: English

We have focused in this section on the English prefield, which have been central to the discussions of the evolution of Germanic IS. Our results showed a striking difference in the evolution of the immediately prefinite positions in V₂ and V₃, where subjects “push out” other constituent types. The absolute initial position in V₃, meanwhile, remains relatively conservative. With regard to the probability of prefinite placement of individual constituent types, we identified a contrast between argumental (direct and indirect objects) vs. non-argumental (adverbials and non-pronominal PPs). We attribute the contrast in argument vs. non-argument placement to the general changes in argument licensing (Kiparsky 1997). We have proposed that prefinite pronouns (personal, temporal or locative) function as ongoing topic situation coordinates, while non-pronominal prefinite locative and temporal adverbials and PPs act as shifters of the respective topic situation coordinate. The raise of prefinite overt subjects and the change in argument licensing can be thus said to have resulted in English changing from a largely “local” anchoring strategy of IS, with prefinite pronouns indicating the ongoing personal, locative, and temporal coordinates, to a “global” anchoring type which has retained the ongoing subject (but not object) personal coordinate, while mostly coordinate *shifts* are indicated for locative, temporal, and personal non-subject coordinates.

5.2 Icelandic

In the wider Germanic context, Icelandic has been traditionally viewed as more conservative compared to its Mainland Scandinavian and West Germanic counterparts (e.g. Milroy & Milroy 1985; Dixon 1997: 9; Trudgill 2004). Nonetheless, various changes have been observed, especially since the release of IcePaHC (Wallenberg et al. 2011), which allows one to track morphosyntactic change across the centuries (e.g. Butt et al. 2014; Kinn et al. 2016; Thráinsson 2017; Booth 2018; Schätzle 2018; Pfaff 2019; Booth & Beck 2021; Wallenberg et al. 2021).

In terms of the syntax-IS interface in Old Icelandic (c. 1150–1350), a good deal of research has focused on the alternation between SUBJ-V order and V-SUBJ order (“narrative inversion V1”), an alternation which is highly characteristic of the early sagas (e.g. Platzack 1985; Hopper 1987; Booth 2018). Kossuth (1980,

1981) represents early work in this area, and observed that while V-SUBJ order is an anaphoric order, signalling an ongoing narrative with no change in participants, SUBJ-V order can occur when a new (non-anaphoric) topical subject is introduced. This is taken up and extended by Booth & Beck (2021), who observe that topical subjects increasingly target the prefield over time, resulting in an increase in SUBJ-V orders at the expense of V1.

With respect to the prefield specifically in Old Icelandic, the received view is indeed that it typically hosts topical constituents (e.g. Kossuth 1978: 455; Faarlund 2004: 231). This extends to pronominal arguments, e.g. (13-a), as well as anaphoric adverbs, e.g. (13-b), which can together be conceived of as topic situation setters.

- (13) a. Rannveig hét önnur dóttir
 Rannveig.NOM was.called second.NOM daughter.NOM
 Ásmundar. **Hana** átti Gamli Þórhallsson
 Ásmundur.GEN she.ACC had Gamli.NOM Þórhallsson
 Vínlendings.
 Vínlendingur.GEN
 ‘Ásmundur’s second daughter was called Rannveig. To her was married Gamli Þórhallsson Vínlendings.’ Old Icelandic, 1310, Grettir.14–15
- b. Eitt sumar kom af Íslandi Oddr... **Þá**
 one.ACC summer.ACC came from Iceland.DAT Oddr.NOM then
 var Haraldur konungur yfir Nórgei.
 was Haraldur.NOM King.NOM over Norway.DAT
 ‘One day Oddr came from Icelandic... At that time Haraldur was King over Norway.’ Old Icelandic, 1275, Morkin.1630–1633

However, as Booth (2021) points out, there is evidence that the Old Icelandic prefield is more diverse than this. For instance, it can also host contrastive expressions, e.g. (14-a), as well as the subject of a presentational sentence, e.g. (14-b) (see discussion in Faarlund 2004: 199).

- (14) a. Context: Then we hallow to God a temple in our hearts...
Níu englasveitum helgum vér musteri ...
 nine.DAT host-angels.DAT hallow.PRS we.NOM temple.ACC
 ‘To nine host angels we (also) hallow a temple...’
 Old Icelandic, 1150, Homiliubok.885 (Booth 2021: 32)
- b. **Björn** er þar ok á því landi.
 bear.NOM be.PRS there also on DEM land
 ‘There is also a bear in that land.’
 Old Icelandic, Kgs 30.10, (Faarlund 2004: 199)

As Booth (2021) notes, in modern Icelandic a bare non-specific indefinite subject in the prefield (cf. Old Icelandic (14-b)) is questionable, or at least not very natural (Thráinsson 2007; Zaenen et al. 2017). This suggests that the range of IS phenomena associated with the Icelandic prefield has become more restricted over time, but this requires closer investigation. Below we examine the frequency of various categories in the prefield over the IcePaHC diachrony (1150–2008).⁹

5.2.1 Subjects

As for English, for the Icelandic data we fit logistic regression models to the binary variable *prefinite/postfinite* within finite main clauses with both nominal (N=20770) and pronominal subjects (N=18763). Both nominal and pronominal subjects manifest an increase in fronting over time, see Fig. 9, and there is a positive and significant effect of time (model with nominal subjects: intercept = -0.2436, $\beta = 0.0004$, $p < .0001$; model with pronominal subjects: intercept = -1.9383, $\beta = 0.0013$, $p < .0001$).

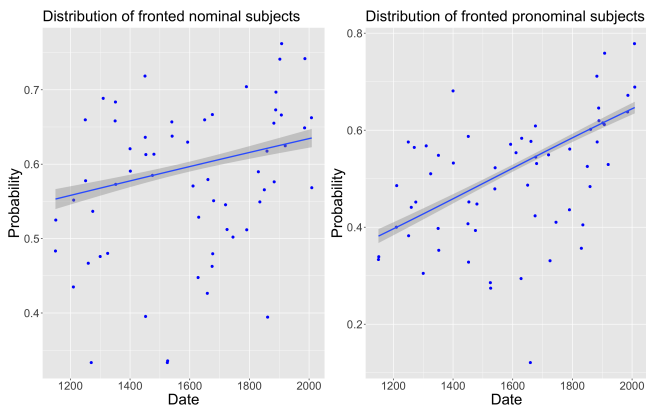


Fig. 9: Logistic regression models of subject fronting for $V=2$ clauses in Icelandic

The observed increase in prefinite subjects is in line with the previous findings in Booth & Beck (2021). However, Booth & Beck (2021) specifically examined the

⁹ In line with the corpus investigation for English (§5.1), for Icelandic we restrict ourselves to clauses annotated with overt non-expletive subjects in IcePaHC, and differentiate between those with nominal and pronominal subjects.

rate of pronominal and definite-marked subjects in the prefield (correlates which they used as proxies for topicality), which they found to increase over time. If we break down nominal subjects into further subtypes and compare the IcePaHC data for Old Icelandic (1150-1350) and modern Icelandic (1900-2008), we can observe that all types of subject are more frequent in the prefield in modern Icelandic compared to Old Icelandic, not just personal pronoun and definite-marked subjects, see Tab. 2. In Old Icelandic, the only type of subject more likely to occur prefinutely rather than postfinutely are subjects which are demonstrative pronouns, e.g. (15). All other subject types are more commonly postfinite than prefinite.

- (15) a. **Petta** fór víða um sveitir.
 DEM.NEUT.NOM travelled widely around districts.ACC
 ‘This spread widely throughout the districts.’
 Old Icelandic, 1350, Finnbogi.628.187
- b. **Sjá** sneri till skálans...
 DEM.MASC.NOM turned to hall.GEN.DEF
 ‘He turned into the hall...’ Old Icelandic, 1310, Grettir.1501

Strikingly, by modern Icelandic, all types of subject are more frequently prefinite than postfinite, including bare (common) nouns, e.g. (16-a), and the indefinite pronoun *maður*, e.g. (16-b).

- (16) a. **Svanahópur** kom norðan að og stefndi til
 swan.group.NOM came from.north PTCL and headed to
 Veiðivatns
 Veiðivatn.GEN
 ‘A group of swans came from the north and headed for Veiðivatn.’
 Modern Icelandic, 1902, Fossar.1459–1460
- b. **Maður** getur veitt skjól og nært það sem
 man.NOM can gave shelter.ACC and nourished DEM.NOM REL
 er veikburða.
 is weak.NOM
 ‘One can give shelter and nourish the weak.’
 Modern Icelandic, 2008, Mamma.920

Category		In prefield		Any position
		<i>n</i>	%	<i>n</i>
1150–1350	personal pronouns	1,439	33.9%	4,245
	demonstrative pronoun	109	59.2%	184
	noun + demonstrative article (<i>sá/þessi</i>)	162	47.5%	341
	noun + suffixed definite article (<i>-inn</i>)	64	34.2%	187
	bare (common) noun	201	42.9%	468
	indefinite pronoun (<i>maður</i>)	30	14.4%	209
1900–2008	personal pronouns	1,770	60.8%	2,910
	demonstrative pronoun	97	76.4%	127
	noun + demonstrative article (<i>sá/þessi</i>)	62	65.3%	95
	noun + suffixed definite article (<i>-inn</i>)	200	71.9%	278
	bare (common) noun	131	61.8%	212
	indefinite pronoun (<i>maður</i>)	59	56.7%	104

Tab. 2: Relative frequencies of types of subject in the prefield in V₌₂ declarative clauses in Old Icelandic (1150–1350) and modern Icelandic (1900–2008) in the IcePaHC corpus

5.2.2 Objects

With respect to direct objects (tagged as NP-OB1) and indirect objects (tagged NP-OB2) in the prefield, which we treat here together, there is a difference between clauses with nominal subjects (N=4889) and clauses with pronominal subjects (N=6765).¹⁰ In the former case, object fronting is diachronically stable, see Fig. 10 (left) but also the results from the model, where the intercept = -2.1488, β = -0.0001, p = 0.7348. In the latter case, there is a significant decrease in fronting probability, see Fig. 10 (right), with an intercept = 0.5411, β = -0.0015, p < .0001. Plausibly, one can relate the particularly strong decrease in object fronting in clauses with pronominal subjects to the fact that fronting of pronominal subjects to the prefield increases more dramatically over time compared to nominal subjects (see Fig. 9 above).

¹⁰ Unlike for English (§5.1), for Icelandic we treat direct and indirect objects together since indirect objects have not undergone such dramatic changes in their morphological expression in the language; in modern Icelandic prepositional indirect objects are only possible to a very limited extent (Thráinsson 2007: 174).

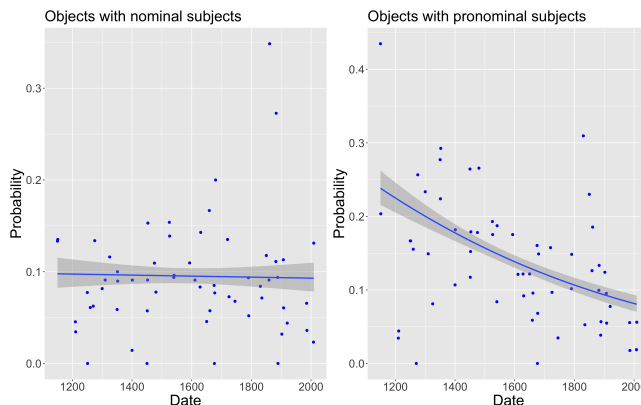


Fig. 10: Logistic regression models of object fronting for $V=2$ clauses in Icelandic

5.2.3 Anaphoric adverbs

We also examine the rate of fronting of the anaphoric adverbs *þá* ('then') and *þar* ('there') over time which we assume to establish the topic situation in relation to the previous utterance, like their English counterparts (see §5.1.4). In clauses with nominal subjects ($N=2671$), there is a statistically significant increase over time, see Fig. 11 (left), with an intercept = -0.130, $\beta = 0.0005$, $p = 0.0033$. In clauses with pronominal subjects ($N=1362$), there is no significant increase, see Fig. 11 (right), with an intercept = -0.217, $\beta = 0.0004$, $p = 0.08$.

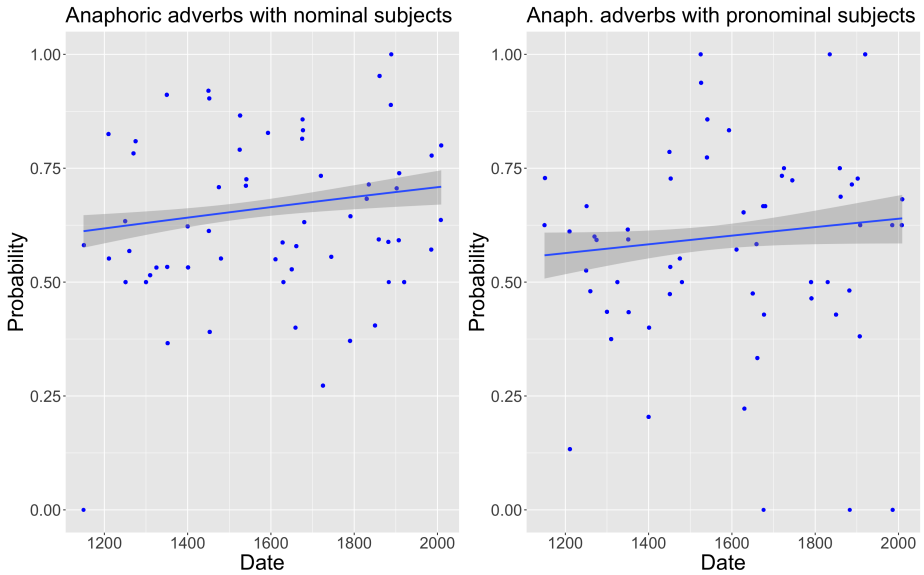


Fig. 11: Logistic regression models of anaphoric adverbs fronting for $V=2$ clauses in Icelandic

5.2.4 Interim summary: Icelandic

In sum, we have observed a significant increase in prefinite subjects in Icelandic over time, similar to English. While in English, object fronting exhibits a significant decline in both clauses with nominal and pronominal subjects, in Icelandic our model detects a significant decrease in object fronting only with pronominal subjects. Consistent with the fact that subject fronting precludes fronting of other constituents in Icelandic (since $V3$ and V -later is ruled out) and that fronting of pronominal subjects increases in Icelandic at a higher rate than fronting of nominal ones, anaphoric adverbs manifest a significant increase in fronting only in the context of nominal subjects. In English, where nominal and pronominal subject fronting increases at very similar rates, we found a slight decrease in adverbial fronting with both subject types.

Our findings concerning an increase in prefinite subjects and (a partial) increase in prefinite anaphoric adverbs are in line with the decrease in $V1$ observed elsewhere for the IcePaHC diachrony (Booth & Beck 2021). In addition, we will later show how this is connected with another change in Icelandic, namely the loss of discourse-partitioning adverbs in the postfinite domain (§6.2).

5.3 High German

The German prefield has been argued to host topic, scene-setting, and contrastive elements. Eckardt & Speyer (2016: 517) claim for Early German that topics and scene-setters occurred in the prefield for pragmatic reasons, whereas contrastive topics occurred there for reasons of prosody. Based on a small textual sample, Speyer (2019) concludes that all three options are present both in Old and in Modern High German. Thus, there are attested Old High German examples where the prefield hosts a topic, e.g. (17-a), examples where it hosts a scene-setter, e.g. (17-b), and examples where it hosts a contrastive expression, e.g. (17-c).

- (17) a. Context: and there was a man whose right hand was withered
 in bihieltun thó thie scribara
 he.ACC watched PTCL DEF.NOM scribes.NOM
 ‘and the scribes watched him’
 Old High German, T 227,10 (Axel 2009: 22)
- b. **In themo tage** giengun zi Imo sadducei
 in thatDAT day.DAT went to he.DAT Sadducees.NOM
 ‘The same day came to him the Sadducees’
 Old High German, T 427,25 (Axel 2007: 133)
- c. **Hérbeste** gáb kót tîe êrâ náls
 autumn.DAT gave God.NOM DEF.ACC honour.ACC not
 temo lénzen
 DEF.DAT spring.DAT
 ‘God has given this award to the autumn, but not to the spring’
 Old High German, N *Boethius* I 36,3 (Axel 2009: 21)

Speyer & Weiß (2018) examine clauses with multiple candidates for the prefield and claim that, in Middle High German, scene-setters are preferred in the prefield over contrastive elements and topics (as in Modern German), while there is no preference ranking evident between fronted topics and contrastive elements where both are present, see the schema in (18) (based on findings in Speyer & Weiß 2018: §5.2.1).

- (18) scene-setter > topic, contrastive element

Additionally, Speyer & Weiß (2018) observe for Middle High German that there is a preference for the prefield to host an adverb which serves as a cohesive device, e.g. (19), which often outranks other prefield candidates (i.e. scene-setters and topic) at this stage of the language. By Early New High German, cohesive devices in the prefield do not play such a central role.

- (19) Und **alsô** hât uns got alle naht und tac in sîner huote
 and so has us God all night and day in his custody
 ‘And consequently, God keeps watch over us all days and nights’
 Middle High German, Berthold I 48, 11–12 (Speyer & Weiß 2018: 69)

Another change is that, by Modern High German, there has been a quantitative shift towards a greater relative frequency of contrastive elements in the prefield (see also Eckardt & Speyer 2016: 515; Hinterhölzl & Petrova 2010; Speyer 2019).

Interestingly, some scholars propose that the prefield in Modern High German can be filled via mechanisms that are not triggered by IS. In fact, as shown by Frey (2004), there are different ways to fill the prefield in Modern High German. The first type of constituent found in the prefield is derived, according to Frey, via a type of fronting that delivers no special pragmatic import to the meaning of the clause, e.g. (20).

- (20) Dem Karl hat das Spiel gut gefallen
 def.DAT Karl has the match good liked
 ‘Karl has liked the match.’ Modern German (Frey 2004: ex. (10-c))

While this kind of placement, traditionally known as *Vorfeldbesetzung*, has often been described as a form of topicalisation (Molnár 1991; Reis 2000), Frey argues against this on the basis that: i) topics can also be found in the middle field, e.g. (21-a) (see also the experimental evidence in Störzer & Stolterfoht 2018), and ii) a quantified non-referential frame-adverbial can occupy the prefield even though it is not topicable, e.g. (21-b).

- (21) a. Ich erzähle dir etwas über Hans. Ich habe gehört,
 I tell you.DAT something on Hans. I have heard
 dass den Hans erfreulicherweise (#den Hans) nächstes
 that def.ACC Hans happily the.ACC Hans next
 Jahr eine polnische Gräfin heiraten wird.
 year a Polish countess marry will.3SG
 ‘I’ll tell you something about Hans. I have heard that next year he
 will marry a Polish countess.’ Modern German, (Frey 2004: 3)
- b. Fast überall ist Fußball sehr beliebt.
 almost everywhere is football very beloved
 ‘Football is loved almost everywhere.’ Modern German, (Frey 2004: 4)

Frey therefore insists on the fact that this type of prefield fill is semantically and pragmatically vacuous with no import to IS. Differently, Eckardt & Speyer (2016) argue that the prefield in German is “a position in the service of information structure” (p. 515), arguing that it can host aboutness and contrastive topics,

and framesetters (see above). The second type of prefield placement that Frey envisages is with discourse-adverbs that express an evaluative mood with regard to the propositional content (e.g. *kein Wunder* “no wonder”, *ein Glück* “a luck”). Frey argues that the prefield can be filled for a third reason: marking a contrast. For example, the prefields in (22-a) and (22-b) must be interpreted with a contrastive meaning, in particular with a contrast on a set of alternatives. This type of focal association can also concern, as exemplified in (22-b), topics, making the prefield eligible for contrastive topics as well.

- (22) a. Grün wird Maria die Tür streichen.
 green will Maria the door paint
 “Maria will paint the door green.” Modern German, (Frey 2004: 21)
- b. Das 1. Halbfinale denke ich, dass sich jeder
 the first semifinal think.1SG I that REFL each.NOM
 Fußballfan anschauen wird
 football.fan watch will.3SG
 “I think that each football fan will watch the first semifinal”
 Modern German, (Frey 2004: 21)

We examine the frequency of different categories in the prefield in the currently available data in the IPCHG corpus (Sapp et al. 2024).¹¹ The current time span of IPCHG does not make it possible to examine the effect of time on the fronting frequencies. We therefore focus on comparing the fronting frequencies of various grammatical categories.

Both in clauses with nominal and pronominal subjects, the category with the strongest preference for the prefield is anaphoric adverbs (*da*, *do*, *dann*, *dar*, *dort*, *hin* and compound forms), see Tab. 3. We can relate this to the previously observed preference for cohesive adverbs to occur in the prefield in earlier stages of High German (see Speyer & Weiß 2018 and discussion above). As for English and Icelandic, we assume that such adverbs set the topic situation in relation to the verification situation of the previous utterance. This is in line with the observations in Demske (2015) and Demske & Wiese (2016), where the prefield in Early New High German shows an increase in resumptive forms, e.g. *do*, *darum*, *damit*, *auch*, *so*.

¹¹ Again, we look separately at $V_{=2}$ matrix declaratives with nominal and pronominal subjects and excluding clauses with expletive subjects. Although it is well known that historical stages of High German also exhibit $V_{=3}$ and $V_{>3}$ clauses (e.g. Walkden 2013; Breitbarth 2023; Catasso & De Bastiani 2024), for reasons of space we do not examine these in this paper.

Nominal subjects are more likely to occur in the prefield than pronominal ones, which contrasts with our results for English, where pronominal subjects are more likely to be fronted than nominal subjects (see §5.1.2). Objects, meanwhile, have a much weaker preference for the prefield and are mostly postfinite, see Tab. 3. We further examine the behaviour of subtypes of subject and object, as we did for Icelandic subjects in §5.2.1. The results are shown in Tab. 4. Both subjects and objects which are expressed as demonstrative pronouns show a very strong fronting preference, although this is stronger for subjects than for objects. Likewise, pronominal subjects, subjects expressed as a noun accompanied by a demonstrative/definite article, and subjects expressed as a noun accompanied by an indefinite article show a stronger preference for the prefield than their object counterparts.

		In prefield		Any position
Category		<i>n</i>	%	<i>n</i>
With nominal subjects	subject	2,413	54.8%	4,398
	object	194	11.8%	1,638
	anaphoric adverb	343	86.6%	396
With pronominal subjects	subject	1,143	40.1%	2,850
	object	221	14.3%	1,536
	anaphoric adverb	189	79.4%	238

Tab. 3: Relative frequencies of constituent types in the prefield in $V_{=2}$ declarative clauses with nominal vs. pronominal subjects in the IPCHG corpus (1360–1650)

The fact that anaphoric adverbs are mostly prefinite rather than postfinite (see Tab. 3) is consistent with the fronting preference for arguments expressed as demonstratives and definite entities (see Table 4) on the assumption that all these elements, when occurring in the prefield, serve to establish the current topic situation by virtue of their semantic composition involving a situational component.

Just as in English, the interpretation of anaphoric temporal adverbs differs depending on whether they are clause-initial or clause-internal. The specific interpretations, however, seem to differ from English. According to Donhauser & Petrova (2009), *tho* (lit. ‘then’) in Old High German, when clause-initial, refers to a time span already established in the preceding section (what is called co-temporal

12 In the IPCHG annotation, there is no clear way to distinguish between demonstrative and definite articles.

Type		In prefield		Any position
		<i>n</i>	%	<i>n</i>
Subjects	personal pronoun	1,143	40.1%	2,850
	demonstrative pronoun	537	95.2%	564
	noun + demonstrative/definite article ¹²	740	52.1%	1,420
	indefinite pronoun (<i>man</i>)	61	27.2%	224
	noun + indefinite article	45	21.4%	210
Objects	personal pronoun	22	2.2%	1,003
	demonstrative pronoun	219	85.2%	257
	noun + demonstrative/definite article	125	18.7%	668
	noun + indefinite article	6	2.2%	274

Tab. 4: Relative frequencies of types of subject and types of object in the prefield in $V_{=2}$ declarative clauses in historical High German in the IPCHG corpus (1360–1650)

in English by Trips & Fuß 2009), e.g. (23-a). In contrast, clause-internal *tho* refers to a novel, indefinite time interval introduced as the topic time (cf. Klein 1994) of a new episode, e.g. (23-b) (see also discussion in Petrova 2011). The latter appears to be effectively a sequential interpretation. If so, strikingly, German shows a mirror picture of the English interpretational pattern, where sequential interpretation is associated with the prefinite position.

- (23) a. **tho** uuas man in hierusalem
then was man in Jerusalem
‘A man lived in Jerusalem at that time’
Latin: *homo erat in hierusalem*
Old High German, T 37,23 (Petrova 2011: 219)
- b. uaas **thó** giuuortan in anderemo sambaztag
was then become in another Sabbath
‘It happened on another Sabbath’
Latin: *Factum est in alio sabbatum autem*
Old High German, T 37,23 (Petrova 2011: 219)

5.4 Low German

In comparison with High German, Low German has been less thoroughly examined diachronically, though there has been a flurry of activity concerning Middle Low German (MLG, c. 1250-1650) syntax in recent years which has demonstrated MLG’s position in its own right within West Germanic (e.g. Mähl 2004, 2014; Tophinke 2009; Petrova 2012; Wallmeier 2015; Merten 2015; Farasyn & Breitbarth

2016; Farasyn 2018; Breitbarth to appear; Ihden 2020). In matrix declaratives, the finite verb is typically in second position; Dreessen & Ihden (2015) observe that, across a subset of four texts from the ReN corpus (ReN-Team 2021), $V_{=2}$ is the dominant verb position in matrix clauses.¹³

According to Petrova (2012: 167), the prefield imposes no restrictions with respect to the IS properties of constituents that can go there. In particular, the prefield in matrix declarative clauses can feature frame-setters, e.g. (24-a), or aboutness topics, e.g. (24-b).

- (24) a. [**Under den alden konningen to Rome**] was en de
among the old kings in Rome was one REL
het Tarquinius
was.called Tarquin
‘Among the old kings in Rome, there was one who was called Tarquin’
- b. [**He**] was de homudigiste man de ie cronen uppe sin
he was the most.malicious man who ever crown on his
hovet satte
head set
‘He was the most malicious man who ever set a crown onto his head.’
Middle Low German, SW 97, 12 (Petrova 2012: 167)

At the same time, Petrova (2012) provides examples where expressions in information focus, e.g. the novel referent introduced in the prefield in (25-a), and expressions in contrastive focus, e.g. (25-b), are attested in the MLG prefield.

- (25) a. [**Enne vrowen**] name he do, diu het Lucretia
INDEF wife took he then she.DET was.called Lucretia
‘Then, he married a woman who was called Lucretia’
Middle Low German, SW 97, 18
- b. [**Van der aller geschlechte**] swige we, und [**van**
of the all descendants keep.silent we and about
Sethis geschlechte] segge we
Seth.GEN descendants tell we
‘About the descendants of the rest, we will keep silent, and about
Seth’s descendants we will continue narrating’
Middle Low German, SW 68, 6, (Petrova 2012: 168)

13 At the same time, it is well known that MLG also exhibits V3 orders in matrix declaratives, see e.g. Petrova (2012); Dreessen & Ihden (2015); Breitbarth (2023); Booth et al. (Forthcoming). For the purposes of this paper, we restrict ourselves to the IS properties of $V_{=2}$ matrix declaratives.

Petrova (2013: 114) also provides examples where “information-structurally vacuous elements” occur in the prefield, including quantified phrases, e.g. (26-a), negative indefinites, e.g. (26-b), and the indefinite subject *men* ‘people’, e.g. (26-c).

- (26) a. [**Iewlich**] lovede de sine
everybody loved DEF.ACC his
‘Everybody adored his (wife)’ Middle Low German, SW 97, 29
- b. [**Neman**] ne dorste brot backen
nobody NEG dared bread bake
‘Nobody dared to bake bread’ Middle Low German, SW 102, 7
- c. [**Men**] vint gescreven oc an enen boke
INDEF.man finds written also in INDEF.ACC book
‘You can also find in a book’ Middle Low German, SW 103, 1
(Petrova 2013: 114)

The studies on the IS properties of the MLG prefield by Petrova (2012, 2013) were conducted manually on a rather limited number of texts and text types, and therefore provide no quantitative information regarding the frequency of different types of constituent in the prefield. In this section, we make use of the newly released MLG component of the CHLG corpus (Booth et al. 2020) to assess this.¹⁴

The findings for the frequency of certain categories in the prefield are shown in Tab. 5. Similarly to the results for High German from the IPCHG corpus (cf. Tab. 3), the broadly contemporaneous MLG German data in Tab. 5 show that anaphoric adverbs (e.g. *dan*, *do* ‘then’, *dar*, ‘there’) are more likely to be prefinite than postfinite, both in clauses with nominal and pronominal subjects, e.g. (27).

- (27) a. **Daer** gaf got adam lijf vnde siele beyde.
there gave got.NOM adam life and soul both
‘There God gave Adam both life and soul.’
Middle Low German, SPIEGHEL.WP.REL.818
- b. **Do** uraghenden se ene: wat bist u den?
then asked they.NOM he.ACC what are you.NOM then
‘Then they asked him: what are you then?’
Middle Low German, BUXTEHUDER.NLG.REL.40

Another parallel with the High German data is that, in the MLG data here, nominal subjects are more frequently prefinite than pronominal subjects, although

¹⁴ We exclude two literary texts from the corpus, *Flos* and *Zeno*, which represent poetic texts where metrical constraints are likely to affect word order. As per the rest of this chapter, our sample is restricted to finite matrix clauses which overt (non-expletive) subjects, and considers clauses with overt nominal and overt pronominal subjects separately.

the difference is small, see Tab. 5. There is no difference in the frequency of prefinite objects between clauses with nominal and pronominal subjects. Breaking down subjects and objects by subtype, it is clear that, as for High German (cf. §5.3), in MLG demonstrative pronouns are more likely to occur prefinally than postfinally, both as subjects and objects, see Tab. 6. In the MLG data – where demonstratives (DDA, DDN) and definite articles (DDIARTA, DDARTN) are distinguishable via the CHLG annotation – this strong fronting preference also extends to subjects and objects expressed as nouns accompanied by demonstrative articles.

		In prefield		Any position
Category		<i>n</i>	%	<i>n</i>
With nominal subjects	subjects	1,184	42.3	2,798
	objects	191	16.7	1,143
	anaphoric adverbs	389	92.6	420
With pronominal subjects	subjects	1,093	38.0	2,879
	objects	231	16.3	1,417
	anaphoric adverbs	153	83.6	183

Tab. 5: Relative frequencies of three prefield categories in $V_{=2}$ declarative clauses with nominal subjects vs. $V_{=2}$ declarative clauses with pronominal subjects in Middle Low German in the CHLG corpus (1279–1580)

		In prefield		Any position
Type		<i>n</i>	%	<i>n</i>
Subjects	personal pronoun	1,093	38.0%	2,879
	demonstrative pronoun	636	97.5%	652
	noun + demonstrative article	59	89.4%	66
	noun + definite article	309	46.9%	659
	indefinite pronoun (<i>men</i>)	20	24.4%	82
	noun + indefinite article	22	20.0%	110
Objects	personal pronoun	9	1.6%	563
	demonstrative pronoun	129	62.6%	206
	noun + demonstrative article	19	61.3%	31
	noun + definite article	33	5.4%	612
	noun + indefinite article	1	0.8%	132

Tab. 6: Relative frequencies of types of subject and types of object in the prefield in $V_{=2}$ declarative clauses in Middle Low German in the CHLG corpus (1279–1580)

6 The postfinite domain: object order and discourse-partitioning

Word order variation in the postfinite domain in Germanic has attracted much attention in the past three decades, in particular in the context of historical stages and the role of information structure as a conditioning factor (e.g. Shannon 2003; Ribbert 2006; van Kemenade & Los 2006; Linde 2009; De Cuypere 2010; Thráinsson 2013; Petrova 2015; Taylor & Pintzuk 2015; Speyer 2018; De Bastiani & Hinterhölzl 2020; Struik & Schoenmakers 2023; Booth Forthcoming). In this section, we focus on two aspects of the postfinite domain which have been particularly thoroughly researched for historical Germanic varieties: (i) the relative order of objects in ditransitives (§6.1) and (ii) the role of anaphoric adverbs as discourse-partitioners (§6.2).

6.1 Object order in ditransitives

The relative order of the two objects in ditransitives, which we will refer to in terms of their thematic roles (Theme, Beneficiary), has been investigated diachronically for a number of Germanic varieties. The overall story borne out from the various studies is the increasing dominance of Beneficiary>Theme order at the expense of Theme>Beneficiary order; see e.g. Polo (2002) on English, van der Feest Viðarsson (2021) on Icelandic and Rauth (2020) on (High and Low) German.¹⁵

Concerning language stages where both Theme>Beneficiary and Beneficiary>Theme order are available, various claims have been made regarding how IS affects the ordering preference. For Old English, for instance, De Cuypere (2010) finds that pronominality, specificity and givenness are relevant conditioning factors for object order in ditransitives. Specifically, pronominal, specific and given objects, whether Themes or Beneficiaries, tend to precede nominal, non-specific and accessible/new objects. However, givenness does not necessarily operate as a conditioning factor in a straightforward way. Petrova (2015) examined Middle Low German ditransitives in terms of the IS status of the objects (given/accessible/brandnew) and found that all IS patterns are attested with both Theme>Beneficiary and Beneficiary>Theme order, and that the principle of Given-before-New is not decisive.

¹⁵ For reasons of space, we leave aside here ditransitives where the Beneficiary is expressed as a PP and focus instead on examples where both Beneficiary and Theme are expressed as noun phrases.

Definiteness has also been identified as a relevant factor. In modern Icelandic – where the canonical order is Beneficiary>Theme and where only a small number of ditransitive verbs still permit Theme>Beneficiary order as a marked option – Theme>Beneficiary order is judged more grammatical when the Theme is definite than when it is indefinite (Thráinsson 2007: 98). Thráinsson (2007: 99, fn. 9) suggests that this effect of definiteness indicates that discourse factors may play a role in the order of objects for those verbs which permit both orders. In this vein, Ottósson (1991: 94) states that, for Theme>Beneficiary order to be possible, the Theme must preferably be unfocused, or at least not more focused than the Beneficiary.

We examine object order in ditransitives in three nearly contemporaneous varieties: (i) Early Icelandic (1150–1450) via IcePaHC (Wallenberg et al. 2011), see Tab. 7; (ii) High German (1360–1650) via the IPCHG (Sapp et al. 2024), see Tab. 8; (ii) Middle Low German (1279–1580) via the CHLG (Booth et al. 2020), see Tab. 9. One generalisation which is borne out by the findings across all three varieties is that, in clauses where one object is pronominal and one object is nominal, there is a very strong preference for the pronominal object to come first, regardless of its thematic role/case-marking. This suggests that information status, and specifically a preference for given-before-new, is one strong factor (among others) in determining the object order in the three varieties.

Clause type		The>Ben		All <i>n</i>
		<i>n</i>	%	
nominal The, nominal Ben	matrix clauses	75	33%	227
	subordinate clauses	40	36%	110
	matrix + subordinate	115	34%	337
pronominal The, pronominal Ben	matrix clauses	2	14%	14
	subordinate clauses	0	0%	13
	matrix + subordinate	2	7%	27
pronominal The, nominal Ben	matrix clauses	8	89%	9
	subordinate clauses	1	100%	1
	matrix + subordinate	9	90%	10
nominal The, pronominal Ben	matrix clauses	5	2%	287
	subordinate clauses	3	2%	123
	matrix + subordinate	8	2%	410

Tab. 7: Relative order of The(me) and Ben(eficiary) objects in the postfinite domain in ditransitive matrix declaratives in Early Icelandic (1150–1450) in IcePaHC

		The>Ben		All
		<i>n</i>	%	<i>n</i>
nominal The, nominal Ben	matrix clauses	45	18%	246
	subordinate clauses	19	14%	316
	matrix + subordinate	64	17%	380
pronominal The, pronominal Ben	matrix clauses	35	66%	53
	subordinate clauses	40	56%	72
	matrix + subordinate	75	60%	125
pronominal The, nominal Ben	matrix clauses	74	93%	80
	subordinate clauses	50	76%	66
	matrix + subordinate	124	85%	146
nominal The, pronominal Ben	matrix clauses	3	1%	224
	subordinate clauses	6	3%	213
	matrix + subordinate	9	2%	437

Tab. 8: Relative order of The(me) and Ben(eficiary) objects in the postfinite domain in ditransitive matrix declaratives in High German in the IPCHG corpus (1360–1650)

		The>Ben		All
		<i>n</i>	%	<i>n</i>
nominal The, nominal Ben	matrix clauses	19	17%	113
	subordinate clauses	3	8%	36
	matrix + subordinate	22	15%	149
pronominal The, pronominal Ben	matrix clauses	10	67%	15
	subordinate clauses	6	35%	17
	matrix + subordinate	16	50%	32
pronominal The, nominal Ben	matrix clauses	15	100%	15
	subordinate clauses	2	33%	6
	matrix + subordinate	17	81%	21
nominal The, pronominal Ben	matrix clauses	0	0%	122
	subordinate clauses	3	8%	38
	matrix + subordinate	3	2%	160

Tab. 9: Relative order of The(me) and Ben(eficiary) objects in the postfinite domain in ditransitive matrix declaratives in Middle Low German in the CHLG corpus (1279–1580)

High German (1360–1650) and Middle Low German show another parallel concerning the difference between clauses where both objects are nominal and clauses where both objects are pronominal. For both varieties, Beneficiary > Theme order is more firmly entrenched in clauses where both objects are nominal than in clauses where both objects are pronominal, cf. Tabs. 8 and 9. Given the broader diachronic trajectory towards Beneficiary > Theme order, clauses with two pronominal objects thus appear to be relatively conservative in this context. The picture for Early Icelandic is slightly different; here, Beneficiary > Theme order is more dominant in clauses with two pronominal objects than in clauses where both objects are nominal, see Tab. 7.

Sensitivity to IS for postfinite word order has been observed for Modern (High) German as well. Before the seminal work by Lenerz (1977), scrambling in the High German middle field was considered free and devoid of any special interpretive effects, since every word order was considered possible. However, Lenerz noticed that some word orders were more marked than others in some contexts, e.g. indefinites and focused constituents are perceived as marked if scrambled.

When it comes to ditransitives, the unmarked word order in the German middle field is Beneficiary > Theme. While this linear order may correspond to either a given-new or new-given IS schema, the marked order Theme > Beneficiary can (but does not have to) be employed when the Theme is given and the Beneficiary is new (in line with the given-before-new preference) (Rauth 2023). Furthermore, Lenerz (1977) suggests that, in cases where both objects are discourse-new or discourse-given, the Theme > Beneficiary order is also possible. Interestingly, these exceptions to the unmarked order were found already in the 11th century in the diachronic study by Rauth (2023). By using logistic regressions to predict the likelihood of the binary variable marked order (Theme > Beneficiary) versus unmarked order (Beneficiary > Theme) on a wide diachronic corpus (11th–20th century for German), Rauth observed that the marked order was favoured if the Theme was discourse-given or salient. In this context, salience was defined as the distance to last mention: the most recently mentioned argument would precede the other, allowing the Theme to precede the Beneficiary.

Another relevant factor observed by Rauth (2023) is definiteness: an indefinite entity tends to follow a definite one. However, Rauth convincingly shows that this effect was not categorical in historical varieties and it is a relatively modern innovation.

Overall, these results are in line with our empirical observations on High and Low German (see Tabs. 8 and 9 respectively); the preference for given > new order potentially overrides the most frequent argument order Beneficiary > Theme. This tendency becomes stronger over time, aligning with morphosyntactic features such

as the expression of definiteness or pronominalization, which often correlate with discourse-given information.

Naturally, it is difficult to ascribe the behaviour of word order to one dimension only, inasmuch as different properties have a high inter-correlation, for example pronouns are generally prosodically weaker than fully-fledged noun phrase. Under this light, we cannot easily single out the factor triggering a different order (see, for example Hinterhölzl (2014) on the interaction between syntax, information-structure and prosody in historical German).

In sum, the previous literature and our data collection show that object order in the High German/Low German postfinite domain is far from “free” but rather governed by a complex interplay between morphology (nominal vs. pronominal arguments), case assignment, predicate type and information structure. This complexity was already detectable in historical stages of German and still applies to modern varieties.

6.2 Anaphoric adverbs as discourse-partitioners

The notion of “information-structural watersheds” goes back to Krivonosov (1977) in relation to the observation that clause-internal discourse particles in modern German segment the postfinite domain into (aboutness) topic and focus fields, e.g. *ja* in (28) (see also Grosz 2016).

- (28) Dann hat [Riko]_{TOPIC} **ja** [eine Frau]_{FOCUS} geküsst
 then has Riko PTCL a woman kissed
 ‘Then Riko has kissed a woman’ Modern German (Grosz 2016: 338)

This is in line with the partition that Frey (2004) had in mind where the top middle field hosts a topical functional projection above sentence adverbials and a right middle field hosts focused or non-focused constituents, provided that they are not topical (see Meinunger 2000 and the problems in Struckmeier 2017).

A similar claim has been made for the anaphoric adverbs *ba/ponne* (‘then’) in Old English in work by van Kemenade and colleagues (e.g. van Kemenade & Los 2006; van Kemenade et al. 2008; van Kemenade & Links 2020). Specifically, clause-internal *ba/ponne* (‘then’) is argued to function as a “discourse-partitioner”, separating presupposed, given information from new, focused information in subordinate clauses, see (29) (adapted from van Kemenade et al. 2008: 10).

- (29) [_{utterance} PRESUPPOSITION TRIGGERS – *ba/ponne* – FOCUS]

The evidence for this claim comes from a series of quantitative corpus findings. For instance, van Kemenade & Los (2006: 232) observe that pronominal subjects in the postfinite domain occur overwhelmingly to the left of *þa/þonne*, e.g. (30); additionally, pronominal objects also regularly appear in this position, as also in the example in (30).

- (30) forþæm he wenð þæt [he] [hi] **þonne** ealle hæbbe
 because he knows that he them then all have
 ‘because he know that he then had them all’
 Old English, *coboeth, Bo*: 24.56.16.1031 (van Kemenade et al. 2008: 12)

By contrast, indefinite nominal subjects are preferred to the right of *þa/þonne* (van Kemenade & Los 2006: 237), e.g. (31); van Kemenade et al. (2008: 13) find that indefinite NPs introducing new discourse entities do not occur to the left of *þa/þonne* in their data.

- (31) Gif **ðonne** [hwelc mon] forbireð his synna
 if then any man forebears his sins
 ‘If anyone then refrains from his sins’
 Old English, *cocura, CP*: 37.265.1.1719 (van Kemenade et al. 2008: 13))

Definite nominal subjects can occur to the left or right of *þa/þonne* but, as van Kemenade et al. (2008: 13) observe, definite nominals occurring in the left position are +anaphoric.

Booth (Forthcoming) has recently shown that a similar IS watershed is in operation in the postfinite domain in Old Icelandic, where clause-internal adverbs including the anaphoric adverbs *þá* (‘then’) and *þar* (‘there’) serve as discourse-partitioners, parallel to Old English *þa/þonne*. Booth shows that different types of subjects show distinct and rather strong preferences in terms of their position with respect to clause-internal *þá/þar*. Pronominal subjects, proper name subjects and definite-marked nominal subjects all favour the position to the left of *þá/þar* (and *nú*, the other adverb which Booth examines), while nominal subjects without definiteness marking and quantified subjects favour the position to the right of the adverb, see Tab. 10. After manual examination of the data, Booth argues that the relevant adverbs serve to mark a rather nuanced IS watershed, which appears to apply specifically to information focus and particular subtypes of topic, see the schema in (32) (adapted from Booth Forthcoming: ex. (37)).¹⁶

¹⁶ “Subsectional topic” as in (32) refers to a topic which is picked out from a previously introduced set of entities, e.g. *Helen and Julie were discussing their plans for the weekend. Julie wanted to go for a bike ride.*

	V-subj- <i>pá</i> / <i>þar</i> / <i>nú</i>		V- <i>pá</i> / <i>þar</i> / <i>nú</i> -subj	
	<i>n</i>	%	<i>n</i>	%
pronominal subject	3,297	99.3%	22	0.7%
proper name subject	859	75.6%	277	24.4%
nominal subject _{+DEF}	78	94.0%	5	6.0 %
nominal subject _{-DEF}	275	21.9%	979	78.1%
quantified subject	10	2.5%	395	97.5%

Tab. 10: Position of different types of (nominative-marked) subject relative to clause-internal *pá*/*þar*/*nú* in matrix declaratives in (M)IcePaHC saga narratives, 1210–1450, (Booth Forthcoming: Table 5)

(32)

V_{finite}	-	$\left\{ \begin{array}{c} \text{continuing topic} \\ \text{shift topic} \\ \text{subsectional topic} \end{array} \right\}$	-	<i>pá</i> / <i>þar</i> / <i>nú</i>	-	$\left\{ \begin{array}{c} \text{information focus} \\ \text{subsectional topic} \end{array} \right\}$
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From a diachronic point of view, both the English and Icelandic IS watersheds are subsequently lost in later periods. van Kemenade & Los (2006) show that, in Middle English, the area to the left of *þa*/*þonne* becomes virtually restricted to subjects, when previously it could host objects, provided the objects were topical/presuppositional. They thus argue for the reanalysis of a topic domain to a subject domain in the left periphery of subordinate clauses, the establishment of SVO order, and with it the loss of *þa*/*þonne* as discourse-partitioners.

Likewise, Booth (Forthcoming), on the basis of IcePaHC data, shows that the relative frequency of clause-internal *þar*/*pá*/*nú* decreases over the course of the history of Icelandic, though it remains a possibility in the latest period (1901–2008). In particular, she shows that in presentational constructions, V1 with clause-internal *þar*/*pá*/*nú*, e.g. (33-a), is virtually in complementary distribution with the prefield expletive *það*, e.g. (33-b); once the expletive becomes established in presentationals, clause-internal *þar*/*pá*/*nú* are scarcely attested.

- (33)
- a.

Verður nú mannfall ógurligt.
becomes NÚ man.loss.NOM terrible.NOM
‘There comes to be a terrible loss of men.’
Icelandic, 1480, Jarlmann.381 (Booth 2018: 260)
- b.

Það komu nokkrir vopnaðir menn af næstu
EXPL came some.NOM armed.NOM men.NOM from next.DAT
bæjum. . .
farms.DAT

‘There came some armed men from the nearby farms. . .’

Modern Icelandic, 2008, Ofsi.634, (Booth Forthcoming: ex. (48-b))

Booth interprets these results to suggest that, in presentationals, the role of marking a thetic, topicless construction shifts diachronically from clause-internal *þar/þá/nú* to clause-initial (prefield) *það*.

In sum, the use of particles as marking an IS watershed in the postfinite domain is common to German, English and Icelandic, but whilst it is a historical feature which has been lost by Modern English and Modern Icelandic (in English lost in the Middle English period; in Icelandic as of c. 1900), in Modern (High) German this phenomenon remains present, as expressed by discourse particles like *ja* (see (28)). On our analysis of anaphoric temporal adverbs as introducers of a temporal extension of the previous verification situation, their “watershed” role can be seen in terms of providing an evaluation domain for structurally lower expressions, while structurally higher expressions have their situation variables set independently.

7 Conclusion

In this chapter, we have approached the diachronic interaction of IS and morphosyntax in Germanic languages via an examination of a number of syntactic alternations, together commonly referred to as “word order flexibility/freedom”, in particular, the prefinite/postfinite alternation and object order in the postfinite domain, as well as constituent placement before/after postfinite particles. The initial motivation for our approach was the assumption about the general economy of expressive means which pushes against many-to-one mappings between language elements (morphemes or configurations) and meanings. The null hypothesis thus consists in attributing to syntactic alternations interpretational effects in terms of the relation between the propositional meaning and context or IS effects.

Using corpus data, we identified constituents with the highest likelihood of fronting to the prefield and considered their semantico-pragmatic make-up in order to draw conclusions about their possible roles in relating the utterance to the context. Our conclusions are in line with much qualitative research suggesting that it is no accident that the most frequently fronted constituents have a pronominal component to them, be it personal, demonstrative or temporal. We have proposed a refinement to the idea that, when fronted, these constituents serve to maintain discourse coherence (Los & van Kemenade 2018; Trips & Fuß 2009 among others) by proposing a situation semantics-based definition of topic whereby the fronting of the constituents in question serves to maintain or change topic situation coordinates,

the running coordinates being intrinsically related to the previous discourse. This can be seen as an extension of the topic situation hypothesis in Frazier & Clifton Jr. (2018). From this perspective, it is the interaction of a basic syntactic hierarchy with the intrinsic semantics of particular morphemes which is employed in managing the relation between the propositional content and the context, rather than special properties of particular syntactic projections. This view is also applicable to literature-based facts about anaphoric particles serving as a “watershed” between presupposition triggers and the rest. We tentatively hypothesise that the particles in question serve to provide an evaluation domain for elements that are not equipped with their own resource situation, which it c-commands. In contrast, the elements whose situation variables are valued independently are placed higher than the particle in question, hence the appearance of a “watershed”. The anaphoric-first (pronominal or anaphorically interpreted full noun phrases) effect has also been identified as governing object order in ditransitives.

Thanks to the large time spans of the available English and Icelandic corpora we have been able to investigate potential changes in fronting frequencies across time. We have established that the likelihood of fronting changes for arguments, while non-arguments front at a frequency that either remains stable or changes slightly. The changes in argument fronting are much more pronounced in English than Icelandic, however, the direction of the change is the same, towards a greater fronting frequency of subjects and a lesser fronting frequency of objects. As the contrast between arguments and non-arguments suggests, rather than being IS-related changes per se, this picture appears to be shaped by the argument licensing component of the grammar, whereby prefinite subject placement (on many accounts SpecTP) gains ground diachronically. Due to the more limited time spans for the historical High and Low German corpora, our study in these cases was limited to a snapshot from a historical window, which revealed similarities with English and Icelandic with respect to the hierarchy of the most frequently fronted elements.

This chapter is clearly far from exhaustive. In particular, we have left aside an analysis of contrastive fronting wherein a prosodic dimension arguably plays an important role and which, unlike non-contrastive fronting, appears to override general argument-licensing mechanisms, as well as a number of possibly IS-related (changing) syntactic configurations such as passivisation, clefts and presentational constructions. The study is of course also not exhaustive cross-linguistically, there being many more modern and historical Germanic varieties which we have not had space to consider. We hope nevertheless to have provided new data, and a coherent enough synthesis of the state-of-the-art which will serve as a basis for further investigations in the field.

References

- Altmann, Gabriel, Haro von Buttlar, Walter Rott & Udo Strauß. 1983. A law of change in language. In Barron Brainerd (ed.), *Historical linguistics*, 104–115. Bochum: Brockmeyer.
- Austin, John L. 1950. Truth. *Philosophical Papers* (3rd ed. 1979) 117–133.
- Axel, Katrin. 2007. *Studies on Old High German syntax: Left sentence periphery, verb placement and verb-second*. Amsterdam: John Benjamins. 10.1075/la.112.
- Axel, Katrin. 2009. The verb-second property in Old High German: Different ways of filling the prefield. In Roland Hinterhölzl & Svetlana Petrova (eds.), *Information structure and language change: New approaches to word order variation in Germanic*, 17–44. Berlin: de Gruyter. 10.1515/9783110216110.1.17.
- Bech, Kristin & Kristine Gunn Eide. 2014a. *Information structure and syntactic change in Germanic and Romance languages*. Amsterdam: John Benjamins. 10.1075/la.213.
- Bech, Kristin & Kristine Gunn Eide. 2014b. Information structure and syntax in Old Germanic and Romance languages. In Kristin Bech & Kristine Gunn Eide (eds.), *Information structure and syntactic change in Germanic and Romance languages*, 1–14. Amsterdam: John Benjamins. 10.1075/la.213.
- Bech, Kristin & Kristine Gunn Eide. 2014c. The ISWOC corpus. Department of Literature, Area Studies and European Languages, University of Oslo. <http://iswoc.github.io>.
- Birner, Betty J. 2004. Discourse functions at the periphery: Noncanonical word order in english. *ZAS Papers in Linguistics* 35(1). 41–62.
- Bohnet, Bernd, Alicia Burga & Leo Wanner. 2013. Towards the annotation of Penn TreeBank with information structure. In Ruslan Mitkov & Jong C. Park (eds.), *Proceedings of the Sixth International Joint Conference on Natural Language Processing*, 1250–1256. Nagoya: Asian Federation of Natural Language Processing.
- Booth, Hannah. 2018. *Clause structure and expletives: syntactic change in Icelandic*. Manchester: University of Manchester dissertation.
- Booth, Hannah. 2021. Revisiting the configurability issue in Old Icelandic. *Glossa: A journal of general linguistics* 6(1). 10.16995/glossa.5804.
- Booth, Hannah. 2022. Desiderata for the annotation of information structure in complex sentences. In Sameer Pradhan & Sandra Kuebler (eds.), *Proceedings of the 16th Linguistic Annotation Workshop (LAW-XVI) within LREC2022*, 31–43. Marseille, France: European Language Resources Association.
- Booth, Hannah. 2024. Beyond given versus new: the proprial article in Old Icelandic. In Kristin Bech & Alex Pfaff (eds.), *Noun phrases in early Germanic languages*, 219–268. Berlin: Language Science Press. 10.5281/zenodo.10641195.
- Booth, Hannah. Forthcoming. Narration and clause-internal adverbs: an information-structural watershed in Old Icelandic. In Ulrike Demske (ed.), *Discourse structure and narration: A diachronic view from Germanic*, xx–xy. Berlin: Language Science Press.
- Booth, Hannah & Christin Beck. 2021. Verb-second and verb-first in the history of Icelandic. *Journal of Historical Syntax* 5(28). 1–53. 10.18148/hs/2021.v5i28.112.
- Booth, Hannah, Anne Breitbarth, Aaron Ecay & Melissa Farasyn. 2020. A Penn-style treebank of Middle Low German. In *Proceedings of the Twelfth International Conference on Language Resources and Evaluation (LREC-2020)*, 766–775. Marseille, France: European Language Resources Association.

- Booth, Hannah, Anne Breitbarth & Melissa Farasyn. Forthcoming. Linke Satzperipherie, Mittelfeld und Nullsubjekte im Mittelniederdeutschen. In Marco Coniglio, Anabel Recker & Heike Sahm (eds.), *Mittelniederdeutsch zwischen Korpuslinguistik und Literaturwissenschaft*, xx–xy. Göttingen: Universitätsverlag.
- Breban, Tine. 2012. Functional shifts and the development of English determiners. In Anneli Meurman-Solin, María José López-Couso & Bettelou Los (eds.), *Information structure and syntactic change in the history of English*, 271–300. Oxford: Oxford University Press. 10.1093/acprof:oso/9780199860210.003.0012.
- Breitbarth, Anne. 2023. V3 after central adverbials in German: Continuity or change? *Journal of Historical Syntax* 7(6–19). 1–47. 10.18148/hs/2023.v7i6-19.138.
- Breitbarth, Anne. to appear. Degrees of integration: Resumption after left-peripheral conditional clauses in Middle Low German. In Karen De Clercq, Liliane Haegeman, Terje Lohndal, Christine Meklenborg Salvesen & Alexandra Simonenko (eds.), *And then there were three. The syntax of V3 adverbial resumption in Germanic and in Romance: a comparative perspective*, Oxford: Oxford University Press.
- Bühler, Karl. 1934. *Theory of language: The representational function of language* [Goodwin, Donald Fraser, trans. 1990]. Amsterdam: John Benjamins. 10.1075/z.164.
- Burridge, Kate. 1993. *Syntactic change in Germanic: Aspects of language change in Germanic with particular reference to Middle Dutch*. Amsterdam: John Benjamins. 10.1075/cilt.89.
- Butt, Miriam, Tina Bögel, Kristina Kotcheva, Christin Schätzle, Christian Rohrdantz, Dominik Sacha, Nicole Dehe & Daniel Keim. 2014. V1 in Icelandic: A multifactorial visualization of historical data. In *Proceedings of VisLR: Visualization as added value in the development, use and evaluation of Language Resources Workshop at the 9th edition of the Language Resources and Evaluation Conference (LREC 2014)*, Reykjavik, Iceland.
- Catasso, Nicholas. 2021. How theoretical is your (historical) syntax? Towards a typology of verb-third in Early Old High German. *The Journal of Comparative Germanic Linguistics* 24. 1–48. 10.1007/s10828-021-09123-7.
- Catasso, Nicholas & Chiara De Bastiani. 2024. The diachrony of V3 in German (and some similarities with Old English). *Italian Journal of Linguistics* 36(1). 1–63. 10.26346/1120-2726-222.
- Chafe, Wallace. 1976. Givenness, contrastiveness, definiteness, subjects, topics, and point of view. In Charles Li (ed.), *Subject and topic*, 25–55. New York: Academic Press.
- Coniglio, Marco, Sonja Linde & Tom Ruetten. 2017. Relative clauses in Old High German: A corpus-based statistical investigation of their syntactic and information-structural properties. *Journal of Germanic Linguistics* 29(2). 101–146. 10.1017/S1470542716000210.
- Cook, Philippa & Felix Bildhauer. 2011. Annotating information structure: the case of topic. In Stefanie Dipper & Heike Zinsmeister (eds.), *Beyond semantics: Corpus-based investigations of pragmatic and discourse phenomena*, 45–56. Bochum: Ruhr-Universität Bochum, Sprachwissenschaftliches Institut.
- Cook, Philippa & Felix Bildhauer. 2013. Identifying “aboutness topics”: two annotation experiments. *Dialogue & Discourse* 4(2). 118–141. 10.5087/dad.2013.206.
- De Bastiani, Chiara & Roland Hinterhölzl. 2020. On the syntax of object pronouns in Old English and Early Middle English. *GlossaÖ a journal of general linguistics* 5(1). 1–29. 10.5334/gjgl.890.

- De Cuypere, Ludovic. 2010. The Old English double object alternation: a discourse-based approach. *Sprachwissenschaft* 35(3). 337–368.
- Demske, Ulrike. 2015. Adverbials at the left periphery: Syntax and information structure in the history of German. Talk presented at the 17th Diachronic Generative Syntax Conference (DiGS), Reykjavik/Iceland.
- Demske, Ulrike & Heike Wiese. 2016. Vorfeld, das. In Carmen Bluhm, Jens Hopperdietzel & Lars Erik Zeige (eds.), *Glossarium amicorum. Festschrift für Karin Donhauser*, 54–70. Berlin: Institut für deutsche Sprache und Linguistik.
- Dixon, R. M. W. 1997. *The rise and fall of languages*. Cambridge: Cambridge University Press. 10.1017/CBO9780511612060.
- Donhauser, Karin & Svetlana Petrova. 2009. Die Rolle des Adverbs *tho* bei der Generalisierung von Verbzweit im Deutschen. In Monika Dannerer, Peter Mauser, Hannes Scheutz & Andreas E. Weiss (eds.), *Gesprochen—geschrieben—gedichtet: Variation und Transformation von Sprache*, 11–24. Berlin: Erich Schmidt Verlag.
- Dreessen, Katharina & Sarah Ihden. 2015. Korpuslinguistische Studien zur mittelniederdeutschen Syntax. *Jahrbuch für Germanistische Sprachgeschichte* 6(1). 249–275. 10.1515/jbgsg-2015-0016.
- Dreschler, Gea. 2014. Tracing overlap in function in historical corpora. In Kristin Bech & Kristine Gunn Eide (eds.), *Information structure and syntactic change in germanic and romance languages*, 111–139. John Benjamins.
- Eckardt, Régine & Augustin Speyer. 2016. Information structure and language change. In Caroline Féry & Shinichiro Ishihara (eds.), *The Oxford handbook of information structure*, 503–522. Oxford: Oxford University Press. 10.1093/oxfordhb/9780199642670.013.23.
- Eckhoff, Hanne Martine, Silvia Luraghi & Marco Passarotti. 2018. The added value of diachronic treebanks for historical linguistics. *Diachronica* 35(3). 297–309. 10.1075/dia.00004.eck.
- Epstein, Richard. 2011. The distal demonstrative as discourse marker in *Beowulf*. *English Language & Linguistics* 15(1). 113–135. 10.1017/S1360674310000304.
- Erteschik-Shir, Nomi. 2007. *Information structure: The syntax-discourse interface*. Oxford: Oxford University Press. 10.1093/oso/9780199262588.001.0001.
- Faarlund, Jan Terje. 1990. *Syntactic change: Toward a theory of historical syntax*. Berlin: de Gruyter. 10.1515/9783110854947.
- Faarlund, Jan Terje. 2004. *The Syntax of Old Norse: with a survey of the inflectional morphology and a complete bibliography*. Oxford: Oxford University Press. 10.1093/acprof:oso/9780199235599.001.0001.
- Farasyn, Melissa. 2018. *Fitting in or standing out? subject agreement phenomena in Middle Low German*. Ghent: Ghent University dissertation.
- Farasyn, Melissa & Anne Breitbarth. 2016. Nullsubjekte im Mittelniederdeutschen. *Beiträge zur Geschichte der deutschen Sprache und Literatur* 138(4). 524–559. 10.1515/bgsl-2016-0040.
- van der Feest Viðarsson, Heimir. 2021. Sögulegar breytingar á orðaröð með tveggja and-laga sögnum í íslensku [Historical word order changes in ditransitive constructions in Icelandic]. *Íslenskt mál og almenn málfræði* 43. 33–69.
- Foster, Robert. 1975. The use of *Pa* in Old and Middle English narratives. *Neuphilologische Mitteilungen* 76(3). 404–414.
- Frazier, Lyn & Charles Clifton Jr. 2018. Topic situations: Coherence by inclusion. *Journal of Memory and Language* 103. 176–190. 10.1016/j.jml.2018.08.006.

- Frey, Werner. 2004. The grammar-pragmatics interface and the German prefield. *Sprache & Pragmatik* 52. 1–39.
- Frey, Werner. 2006a. Contrast and movement to the German prefield. In Valéria Molnár & Susanne Winkler (eds.), *The architecture of focus*, 235–264. Berlin: de Gruyter. 10.1515/9783110922011.235.
- Frey, Werner. 2006b. How to get an object-es into the German prefield. In Patrick Brandt & Eric Fuss (eds.), *Form, structure, and grammar – a festschrift presented to Günther Grewendorf on occasion of his 60th birthday*, 159–185. Berlin: Akademie-Verlag.
- Fuß, Eric. 2008. Word order and language change: On the interface between syntax and morphology. Frankfurt am Main: Goethe University Frankfurt, Habilitation.
- Féry, Caroline & Shinichiro Ishihara. 2016. Introduction. In Caroline Féry & Shinichiro Ishihara (eds.), *The Oxford handbook of information structure*, 1–15. Oxford: Oxford University Press. 10.1093/oxfordhb/9780199642670.013.43.
- Grosz, Patrick G. 2016. Information structure and discourse particles. In Caroline Féry & Shinichiro Ishihara (eds.), *The Oxford handbook of information structure*, 336–358. Oxford: Oxford University Press. 10.1093/oxfordhb/9780199642670.013.36.
- Haeberli, Eric. 2007. The development of subject-verb inversion in middle english and the role of language contact. *Generative Grammar in Geneva* 5. 15–33.
- Haug, Dag T. T. 2015. Treebanks in historical linguistic research. In Carlotta Viti (ed.), *Perspectives on historical syntax*, 185–202. Amsterdam: John Benjamins. 10.1075/slcs.169.
- Haug, Dag T.T., Hanne Martine Eckhoff & Eirik Welo. 2014. The theoretical foundations of givenness annotation. In Kristin Bech & Kristine Gunn Eide (eds.), *Information structure and syntactic change in Germanic and Romance languages*, 17–52. Amsterdam: John Benjamins. 10.1075/la.213.02hau.
- Hinterhölzl, Roland. 2014. On the interaction between syntax, prosody and information structure: An interface approach to word order developments in germanic. In *Information structure and syntactic change in germanic and romance languages*, 341–376. John Benjamins Publishing Company.
- Hinterhölzl, Roland & Svetlana Petrova. 2009. *Information structure and language change: New approaches to word order variation in Germanic*. Berlin: de Gruyter. 10.1515/9783110216110.
- Hinterhölzl, Roland & Svetlana Petrova. 2010. From V1 to V2 in West Germanic. *Lingua* 120(2). 315–328. 10.1016/j.lingua.2008.10.007.
- Hirschberg, Julia. 1991. *A theory of scalar implicature*. New York: Garland.
- Hopper, Paul J. 1987. Stability and change in VN/NV alternating languages: a study in pragmatics and linguistic typology. In Jef Verschueren & Marcella Bertucelli-Papi (eds.), *The Pragmatic Perspective*, 455–476. Amsterdam: John Benjamins.
- Hróarsdóttir, Þorbjörg. 2008. Types of DPs in OV order. *Studia Linguistica* 62(3). 261–286. 10.1111/j.1467-9582.2008.00151.x.
- Ihden, Sarah. 2020. *Relativsätze im Mittelniederdeutschen: Korpuslinguistische Untersuchungen zu Struktur und Gebrauch*. Berlin: de Gruyter. 10.1515/9783110678468.
- Jacobs, Joachim. 2001. The dimensions of topic-comment. *Linguistics* 39(4). 641–682.
- van Kemenade, Ans & Meta Links. 2020. Discourse particles in early English: Clause structure, pragmatics and discourse management. *Glossa: a journal of general linguistics* 5(1). 10.5334/gjgl.1020.

- van Kemenade, Ans & Bettelou Los. 2006. Discourse adverbs and clausal syntax in Old and Middle English. In Ans van Kemenade & Bettelou Los (eds.), *The handbook of the history of English*, 224–248. Oxford: Wiley Blackwell. 10.1002/9780470757048.ch10.
- van Kemenade, Ans, Tanja Milicev & R. Harald Baayen. 2008. The balance between syntax and discourse in Old English. In Maurizio Gotti, Marina Dossena & Richard Dury (eds.), *English Historical Linguistics 2006: Selected papers from the fourteenth International Conference on English Historical Linguistics (ICEHL 14), Bergamo, 21–25 August 2006*, 3–21. Amsterdam: John Benjamins. 10.1075/cilt.295.04kem.
- Kemenade, Ans van. 1997. Negative-initial sentences in Old and Middle English. *Studia Anglica Posnaniensa* (31). 91–104.
- Kerz, Elma. 2012. The role of genre in information structuring in English. *Belgian Journal of Linguistics* 26(1). 143–159. 10.1075/bjl.26.06ker.
- Kinn, Kari, Kristian A Rusten & George Walkden. 2016. Null subjects in early Icelandic. *Journal of Germanic Linguistics* 28(1). 31–78. 10.1017/S1470542715000136.
- Kiparsky, Paul. 1997. The rise of positional licensing. In Ans van Kemenade & Nigel Vincent (eds.), *Parameters of Morphosyntactic Change*, 460–494. Cambridge: Cambridge University Press.
- Kiss, Katalin É (ed.). 1995. *Discourse configurational languages*. Oxford: Oxford University Press. 10.1093/oso/9780195088335.001.0001.
- Klein, Wolfgang. 1994. *Time in language*. London: Routledge. 10.4324/9781315003801.
- Komen, Erwin. 2011. Cesax: Coreference editor for syntactically annotated XML corpora. Nijmegen: Radboud University Nijmegen. <https://cls.ru.nl/staff/ekomen/software/Cesax/>.
- Komen, Erwin R. 2012. Coreferenced corpora for information structure research. *Studies in variation, contacts and change in English* 10. <http://www.helsinki.fi/varieng/series/volumes/10/komen>.
- Komen, Erwin R. 2013a. *Finding focus: a study of the historical development of focus in English*. Utrecht: Netherlands Graduate School of Linguistics (LOT).
- Komen, Erwin R. 2013b. Predicting referential states using enriched texts. In Francesco Mambrini, Marco Passarotti & Caroline Sporleder (eds.), *Proceedings of the Third Workshop on Annotation of Corpora for Research in the Humanities (ACRH-3)*, 49–60. Sofia: The Institute of Information and Communication Technologies, Bulgarian Academy of Sciences.
- Komen, Erwin R., Rosanne Hebing, Ans Van Kemenade & Bettelou Los. 2014. Quantifying information structure change in English. In Kristin Bech & Kristine Gunn Eide (eds.), *Information structure and syntactic change in Germanic and Romance languages*, 81–110. Amsterdam: John Benjamins. 10.1075/la.213.04kom.
- Kossuth, Karen C. 1978. Icelandic word order: In support of drift as a diachronic principle specific to language families. In *Proceedings of the Fourth Annual Meeting of the Berkeley Linguistics Society*, 446–457. 10.3765/bls.v4i0.2197.
- Kossuth, Karen C. 1980. The linguistic basis of saga structure: Toward a syntax of narrative. *Arkiv för nordisk filologi* 95. 126–141.
- Kossuth, Karen C. 1981. Unmarked definite NPs and referential cohesion in Old Icelandic narrative. *Islensk mál* 3. 85–100.
- Kratzer, Angelika. 2023. Situations in Natural Language Semantics. In Edward N. Zalta & Uri Nodelman (eds.), *The Stanford encyclopedia of philosophy*, Metaphysics Research Lab, Stanford University Fall 2023 edn.

- Krifka, Manfred. 2008. Basic notions of information structure. *Acta Linguistica Hungarica* 55(3/4). 243–276. 10.1556/ALing.55.2008.3-4.2.
- Krivososov, Aleksej. 1977. Deutsche Modalpartikeln im System der unflektierten Wortklassen. In Harald Weydt (ed.), *Aspekte der Modalpartikeln: Studien zur deutschen Abtönung*, 176–216. Tübingen: Niemeyer.
- Kroch, Anthony, Beatrice Santorini & Lauren Delfs. 2004. The Penn-Helsinki Parsed Corpus of Early Modern English (PPCEME). Department of Linguistics, University of Pennsylvania. <https://www.ling.upenn.edu/ppche/ppche-release-2016/PPCEME-RELEASE-3>.
- Kroch, Anthony, Beatrice Santorini & C.E.A. Diertani. 2016. The Penn Parsed Corpus of Modern British English (PPCMBE2). Department of Linguistics, University of Pennsylvania. <https://www.ling.upenn.edu/ppche/ppche-release-2016/PPCMBE2-RELEASE-1>.
- Kroch, Anthony, Ann Taylor & Don Ringe. 2008. The middle english verb-second constraint: A case study in language contact and language change. In *Textual parameters in older languages*, 353–392. John Benjamins Publishing Company.
- Kroch, Anthony, Ann Taylor & Beatrice Santorini. 2000. The Penn-Helsinki Parsed Corpus of Middle English (PPCME2). Department of Linguistics, University of Pennsylvania. CD-ROM, second edition. <https://www.ling.upenn.edu/ppche/ppche-release-2016/PPCME2-RELEASE-4>.
- Kroch, Antony. 1989. Reflexes of grammar in patterns of language change. *Language Variation and Change* 1(3). 199–244. 10.1017/S0954394500000168.
- Kroeger, Paul. 2017. Basic concepts in information structure: Topic, focus, and contrast. *GIALens* 11(1). 12.
- Labov, William. 1994. *Principles of linguistic change. volume i: Internal factors*. Oxford: Blackwell.
- Lambrecht, Knud. 1994. *Information structure and sentence form: Topic, focus, and the mental representations of discourse referents*. Cambridge: Cambridge University Press. 10.1017/CBO9780511620607.
- Lenerz, Jürgen. 1977. *Zur Abfolge nominaler Satzglieder im Deutschen*. Tübingen: Gunter Narr.
- Light, Caitlin. 2011. The information structure of subject extraposition in Early New High German. In Stefan Müller (ed.), *Proceedings of the HPSG 2011 Conference*, 314–326. Stanford, CA: CSLI Publications.
- Light, Caitlin. 2012. *The syntax and pragmatics of fronting in germanic*. Philadelphia, PA: University of Pennsylvania dissertation.
- Linde, Sonja. 2009. Aspects of word order and information structure in Old Saxon. In Roland Hinterhölzl & Svetlana Petrova (eds.), *Information structure and language change: New approaches to word order variation in Germanic*, 367–389. Berlin: de Gruyter. 10.1515/9783110216110.3.367.
- Links, Meta. 2018. *Correlative constructions in earlier english: Clause structure and discourse organization*. Utrecht: LOT.
- Los, Bettelou & Gea Dreschler. 2012. The loss of local anchoring: From adverbial local anchors to permissive subjects. In Terttu Nevalainen & Elizabeth Closs Traugott (eds.), *The oxford handbook of the history of english*, 859–872. OUP.
- Los, Bettelou, Gea Dreschler, Ans van Kemenade, Erwin Komen & Stefano Coretta. 2023. The decline of local anchoring: A quantitative investigation. *English Language & Linguistics* 27(2). 345–372. 10.1017/S1360674323000047.

- Los, Bettelou & Ans van Kemenade. 2018. Syntax and the morphology of deixis: The loss of demonstratives and paratactic clause linking. In Bettelou Los, Ans van Kemenade, Marco Coniglio & Andrew Murphy (eds.), *Atypical demonstratives*, 127–157. Berlin: Mouton de Gruyter. 10.1515/9783110560299-005.
- Lüdeling, Anke, Julia Ritz, Manfred Stede & Amir Zeldes. 2016. Corpus linguistics and information structure research. In Caroline Féry & Shinichiro Ishihara (eds.), *The Oxford handbook of information structure*, 599–620. Oxford: Oxford University Press. 10.1093/oxfordhb/9780199642670.013.013.
- Mähl, Stefan. 2004. *Studien zum mittelniederdeutschen Adverb*. Köln/Wien: Böhlau.
- Mähl, Stefan. 2014. *Mehrgliedrige Verbalkomplexe im Mittelniederdeutschen: ein Beitrag zu einer historischen Syntax des Deutschen*. Köln/Wien: Böhlau.
- Matić, Dejan. 2015. Information structure in linguistics 12. 95–99. 10.1016/B978-0-08-097086-8.53013-X.
- Matić, Dejan & Daniel Wedgwood. 2013. The meanings of focus: The significance of an interpretation-based category in cross-linguistic analysis. *Journal of Linguistics* 49(1). 127–163. 10.1017/S0022226712000345.
- McNally, Louise. 1998. On the linguistic encoding of information packaging instructions. In Peter Culicover & Louise McNally (eds.), *The limits of syntax*, 161–183. New York: Academic Press. 10.1163/9789004373167_007.
- Meinunger, André. 2000. *Syntactic aspects of topic and comment*. Amsterdam: John Benjamins. 10.1075/la.38.
- Merten, Marie-Luis. 2015. Sprachausbau des Mittelniederdeutschen im Kontext rechtssprachlicher Praktiken. Konstruktionsgrammatik meets Kulturanalyse. *Niederdeutsches Jahrbuch* 138. 27–51.
- Merten, Marie-Luis & Nina Seemann. 2018. Analyzing constructional change: Linguistic annotation and sources of uncertainty. In *Proceedings of the sixth international conference on technological ecosystems for enhancing multiculturalism TEEM'18*, 819–825. New York, NY, USA: Association for Computing Machinery. 10.1145/3284179.3284320.
- Meurman-Solin, Anneli, Maria Jose Lopez-Couso & Bettelou Los. 2012. *Information structure and syntactic change in the history of English*. Oxford: Oxford University Press. 10.1093/acprof:oso/9780199860210.001.0001.
- Milroy, James & Lesley Milroy. 1985. Linguistic change, social network and speaker innovation. *Journal of linguistics* 21(2). 339–384. 10.1017/S0022226700010306.
- Molnár, Valéria. 1991. *Das TOPIK im Deutschen und Ungarischen*. Stockholm: Almqvist Wiksell International.
- Öhl, Peter. 2010. Formal and functional constraints on constituent order and their universality. In Carsten Breul & Edward Göbbel (eds.), *Comparative and contrastive studies of information structure*, 231–275. Amsterdam: John Benjamins. 10.1075/la.165.09ohl.
- Ottósson, Kjartan G. 1991. Icelandic double objects as small clauses. *Working Papers in Scandinavian Syntax* 48. 77–97.
- Petrova, Svetlana. 2011. Modeling word order variation in discourse: On the pragmatic properties of VS order in Old High German. *Oslo Studies in Language* 3(3). 209–228. 10.5617/osl.46.
- Petrova, Svetlana. 2012. Multiple XP-fronting in Middle Low German root clauses. *The Journal of Comparative Germanic Linguistics* 15. 157–188. 10.1007/s10828-012-9050-y.
- Petrova, Svetlana. 2013. *The syntax of Middle Low German*. Habilitation, Humboldt-Universität zu Berlin.

- Petrova, Svetlana. 2015. Free word order in Germanic: Insights from object order in Middle Low German. *Linguistische Berichte* 2015(244). 355–382.
- Petrova, Svetlana & Michael Solf. 2009. On the methods of information-structural analysis in historical texts: A case study on Old High German. In Roland Hinterhölzl & Svetlana Petrova (eds.), *Information structure and language change: New approaches to word order variation in Germanic*, 121–160. Berlin: de Gruyter. 10.1515/9783110216110.2.121.
- Petrova, Svetlana, Michael Solf, Julia Ritz, Christian Chiarcos & Amir Zeldes. 2009. Building and using a richly annotated interlinear diachronic corpus: The case of Old High German Tatian. *Traitement automatique des Langues* 50(2). 47–71.
- Pfaff, Alexander. 2019. Reunited after 1000 years: The development of definite articles in Icelandic. *Nordic Journal of Linguistics* 42(2). 165–207. 10.1017/S0332586519000155.
- Pintzuk, Susan, Ann Taylor & Anthony Warner. 2017. Corpora and quantitative methods. In Adam Ledgeway & Ian Roberts (eds.), *The Cambridge handbook of historical syntax*, 218–240. Cambridge: Cambridge University Press. 10.1017/9781107279070.012.
- Platzack, Christer. 1985. Narrative inversion in Old Icelandic. *Íslenskt mál* 7. 127–144.
- Polo, Chiara. 2002. Double objects and morphological triggers for syntactic case. In David W. Lightfoot (ed.), *Syntactic effects of morphological change*, 124–142. Oxford University Press Oxford. 10.1093/acprof:oso/9780199250691.003.0007.
- Postma, Gertjan. 2010. The impact of failed changes. In Anne Breitbarth, Christopher Lucas, Sheila Watts & David Willis (eds.), *Continuity and change in grammar*, 269–302. Amsterdam: John Benjamins. 10.1075/la.159.13pos.
- R Core Team. 2013. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing Vienna, Austria. <http://www.R-project.org/>.
- Randall, Beth, Anthony Kroch & Beatrice Santorini. 2004. CorpusSearch 2 user's guide. <https://github.com/beatrice57/CorpusSearch>.
- Rauth, Philipp. 2020. *Ditransitive Konstruktionen im Deutschen: Geschichte und Steuerung der Objektfolge im Mittelfeld*. Tübingen: Stauffenburg.
- Rauth, Philipp. 2023. Ditransitive constructions in the history of German. In Eva Zehentner, Melanie Röthlisberger & Timothy Coleman (eds.), *Ditransitives in Germanic languages*, 80–114. Amsterdam: John Benjamins. 10.1075/sigl.7.03rau.
- Reinhart, Tanya. 1981. Pragmatics and linguistics: An analysis of sentence topics. *Philosophica* 27.
- Reis, Marga. 2000. Anmerkungen zu verb-erst-satz-typen im deutschen. *Deutsche Grammatik in Theorie und Praxis* 215–227.
- ReN-Team. 2021. Reference Corpus Middle Low German/Low Rhenish (1200–1650); Referenzkorpus Mittelniederdeutsch/Niederrheinisch (1200–1650) (Version 1.1) [Data set]. 10.25592/uhhfdm.9195.
- Ribbert, Anne. 2006. Object preposing in 15th century Drenthe. *Linguistics in the Netherlands* 23(1). 174–185. 10.1075/avt.23.18rib.
- Rooth, Mats. 1992. A theory of focus interpretation. *Natural language semantics* 1(1). 75–116. 10.1007/BF02342617.
- Sapp, Christopher D, Elliott Evans, Rex Sprouse & Daniel Dakota. 2024. Introducing a Parsed Corpus of Historical High German. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, 9224–9233.
- Schätzle, Christin. 2018. *Dative subjects: Historical change visualized*. Konstanz: University of Konstanz dissertation. <http://kops.uni-konstanz.de/handle/123456789/45157>.

- Schätzle, Christin & Hannah Booth. 2019. DiaHClust: an iterative hierarchical clustering approach for identifying stages in language change. In *Proceedings of the 1st International Workshop on Computational Approaches to Historical Language Change*, 126–135. Florence, Italy: Association for Computational Linguistics. <https://www.aclweb.org/anthology/W19-4716>.
- Schiffirin, Deborah. 1992. Anaphoric then: Aspectual, textual, and epistemic meaning. *Linguistics* (30). 753–792.
- Schwarz, Florian. 2019. Definites, domain restriction, and discourse structure in online processing. In K. Carlson, C. Clifton Jr. & J. Fodor (eds.), *Grammatical approaches to language processing: Essays in honor of Lyn Frazier*, 187–208. Heidelberg and New York: Springer.
- Shannon, Thomas F. 2003. Drift in Dutch: Fleshing out the factors of change. In Arie Verhagen & Jeroen van de Weijer (eds.), *Usage-based approaches to Dutch: Lexicon, grammar, discourse*, 131–174. Utrecht: LOT, Netherlands Graduate School of Linguistics.
- Song, Sanghoun. 2017. *Modeling information structure in a cross-linguistic perspective*. Berlin: Language Science Press. 10.5281/zenodo.818365.
- Speyer, Augustin. 2010. *Topicalization and stress clash avoidance in the history of english*. Berlin: de Gruyter. 10.1515/9783110220247.
- Speyer, Augustin. 2018. Serialization of full noun phrases in the history of German. In Agnes Jäger, Gisella Ferraresi & Helmut Weiß (eds.), *Clause structure and word order in the history of German*, 155–180. Oxford: Oxford University Press. 10.1093/oso/9780198813545.001.0001.
- Speyer, Augustin. 2019. Topichood and the margins of the German clause from a historical perspective. In Valéria Molnár, Verner Egerland & Susanne Winkler (eds.), *Architecture of topic*, 337–371. Berlin: de Gruyter. 10.1515/9781501504488-012.
- Speyer, Augustin & Helmut Weiß. 2018. The prefield after the Old High German period. In Agnes Jäger, Gisella Ferraresi & Helmut Weiß (eds.), *Clause structure and word order in the history of German*, 64–81. Oxford: Oxford University Press. 10.1093/oso/9780198813545.003.0005.
- Stalnaker, Robert C. 1978. Assertion. In Peter Cole (ed.), *Syntax and semantics* 9, 315–332. Leiden: Brill.
- Stevens, Jon & Caitlin Light. 2013. The pragmatics of direct object fronting in historical English. *University of Pennsylvania Working Papers in Linguistics* 19(1). 23.
- Störzer, Melanie & Britta Stalterfoht. 2018. Is German discourse-configurational? experimental evidence for a topic position. *Glossa: a journal of general linguistics* 3(1).
- Strawson, Peter Frederick. 1964. Identifying reference and truth values. *Theoria* 30.
- Struckmeier, Volker. 2017. Against information structure heads: A relational analysis of German scrambling. *Glossa: a journal of general linguistics* 2(1). 10.5334/gjgl.56.
- Struik, Tara. 2022a. *Information structure triggers for word order variation and change: the OV/VO alternation in the West-Germanic languages*. Amsterdam: LOT.
- Struik, Tara. 2022b. OV/VO variation and information structure in Old Saxon and Middle Low German. *Journal of Historical Syntax* 6(1). 1–36. 10.18148/hs/2022.v6i1.127.
- Struik, Tara & Ans van Kemenade. 2020. On the givenness of OV word order: A (re)examination of OV/VO variation in Old English. *English Language & Linguistics* 24(1). 1–22. 10.1017/S1360674318000187.

- Struik, Tara & Gert-Jan Schoenmakers. 2023. When information structure exploits syntax: The relation between the loss of VO and scrambling in Dutch. *Journal of Linguistics* 59(3). 655–690. 10.1017/S0022226722000172.
- Surányi, Balázs. 2016. Discourse-configurationality. In Caroline Féry & Shinichiro Ishihara (eds.), *The Oxford handbook of information structure*, 422–440. Oxford: Oxford University Press. 10.1093/oxfordhb/9780199642670.013.37.
- Taylor, Ann. 2020. Treebanks in historical syntax. *Annual Review of Linguistics* 6. 195–212. 10.1146/annurev-linguistics-011619-030515.
- Taylor, Ann & Susan Pintzuk. 2012. Rethinking the OV/VO alternation in Old English: The effect of complexity, grammatical weight, and information status. In Terttu Nevalainen & Elizabeth Closs Traugott (eds.), *The Oxford handbook of the history of English*, 835–845. Oxford: Oxford University Press. 10.1093/oxfordhb/9780199922765.013.0068.
- Taylor, Ann & Susan Pintzuk. 2014. Testing the theory: Information structure in Old English. In Kristin Bech & Kristine Gunn Eide (eds.), *Information structure and syntactic change in Germanic and Romance languages*, 53–77. Amsterdam: John Benjamins. 10.1075/la.213.
- Taylor, Ann & Susan Pintzuk. 2015. Verb order, object position and information status in Old English. In Theresa Biberauer & George Walkden (eds.), *Syntax over time: Lexical, morphological, and information-structural interactions*, 318–335. Oxford: Oxford University Press. 10.1093/acprof:oso/9780199687923.003.0019.
- Taylor, Ann, Anthony Warner, Susan Pintzuk & Frank Beths. 2003. The York-Toronto-Helsinki Parsed Corpus of Old English Prose (YCOE). Department of Linguistics, University of York. Oxford Text Archive, first edition. <http://www-users.york.ac.uk/~lang22/YcoeHome1.htm>.
- Thompson, Ellen. 1999. The temporal structure of discourse: the syntax and semantics of temporal then. *Natural Language & Linguistic Theory* 17(1). 123–160.
- Thráinsson, Höskuldur. 2007. *The syntax of Icelandic*. Cambridge: Cambridge University Press. 10.1017/CBO9780511619441.
- Thráinsson, Höskuldur. 2013. Full NP object shift: The Old Norse puzzle and the Faroese puzzle revisited. *Nordic Journal of Linguistics* 36(2). 153–186. 10.1017/S033258651300022X.
- Thráinsson, Höskuldur. 2017. Developing a new perfect: the rise of the Icelandic *vera búinn að*-perfect. *Acta Linguistica Hafniensia* 49(2). 118–142. 10.1080/03740463.2017.1357267.
- Tonhauser, Judith. 2012. Diagnosing (not-) at-issue content. *Proceedings of Semantics of Under-represented Languages of the Americas (SULA)* 6. 239–254.
- Tophinke, Doris. 2009. Vom Vorlesetext zum Lesetext: Zur Syntax mittelniederdeutscher Rechtsverordnungen im Spätmittelalter. In Angelika Linke & Helmuth Feilke (eds.), *Oberfläche und Performanz: Untersuchungen zur Sprache als dynamische Gestalt*, 161–186. Niemeyer.
- Trips, Carola & Eric Fuß. 2009. The syntax and semantics of the temporal anaphor “then” in Old and Middle English. In Artemis Alexiadou, Jorge Hankamer, Thomas McFadden, Justin Nuger & Florian Schäfer (eds.), *Advances in comparative Germanic syntax*, 171–196. Amsterdam: John Benjamins. 10.1075/la.141.08the.
- Trudgill, Peter. 2004. Linguistic and social typology. In J. K. Chambers, Peter Trudgill & Natalie Schilling-Estes (eds.), *The handbook of language variation and change*, 707–728. Oxford: Wiley-Blackwell. 10.1002/9780470756591.ch27.

- Vallduví, Enric. 1992. *The informational component*. New York, NY: Garland.
- Vallduví, Enric & Elisabet Engdahl. 1996. The linguistic realization of information packaging. *Linguistics* 34(3). 459–19. 10.1515/ling.1996.34.3.459.
- Van Kemenade, Ans & Bettelou Los. 2009. *The handbook of the history of english*, vol. 39. Oxford: Blackwell.
- Walkden, George. 2013. Verb-third in early West Germanic: A comparative perspective. In Theresa Biberauer & George Walkden (eds.), *Syntax over time: Lexical, morphological and information-structural interactions*, 236–248. Oxford: Oxford University Press. 10.1093/acprof:oso/9780199687923.003.0014.
- Walkden, George. 2014. Object position and Heavy NP Shift in Old Saxon and beyond. In Kristin Bech & Kristine Gunn Eide (eds.), *Information structure and syntactic change in Germanic and Romance languages*, 313–340. Amsterdam: John Benjamins. 10.1075/la.213.12wal.
- Walkden, George & Kristian Rusten. 2017. Null subjects in Middle English. *English Language and Linguistics* 21(3). 439–473.
- Wallage, Phillip. 2016. Identifying the role of pragmatic activation in changes to the expression of English negation. In Pierre Larrivée & Chungmin Lee (eds.), *Negation and polarity: Experimental perspectives*, 199–227. Leiden: Springer. 10.1007/978-3-319-17464-8_9.
- Wallage, Phillip. 2017. *Negation in Early English: Grammatical and functional change*. Cambridge: Cambridge University Press.
- Wallenberg, Joel C, Rachael Bailes, Christine Cuskley & Anton Karl Ingason. 2021. Smooth signals and syntactic change. *Languages* 6(2). 1–20. 10.3390/languages6020060.
- Wallenberg, Joel C., Anton Karl Ingason, Einar Freyr Sigurðsson & Eiríkur Rögnvaldsson. 2011. Icelandic Parsed Historical Corpus (IcePaHC), version 0.9. [http://linguist.is/icelandic_treebank/Icelandic_Parsed_Historical_Corpus_\(IcePaHC\)](http://linguist.is/icelandic_treebank/Icelandic_Parsed_Historical_Corpus_(IcePaHC)).
- Wallmeier, Nadine. 2015. Konditionale Adverbialsätze und konkurrierende Konstruktionen in mittelniederdeutschen Rechtstexten. *Niederdeutsches Jahrbuch* 138. 7–26.
- Weerman, Fred. 1997. On the relations between morphological and syntactic case. In Ans van Kemenade & Nigel Vincent (eds.), *Parameters of morphosyntactic change*, 427–459. Cambridge: John Benjamins.
- Zaenen, Annie, Elisabet Engdahl & Joan Maling. 2017. Subject properties in presentational sentences in Icelandic and Swedish. In Victoria Rosén & Koenraad de Smedt (eds.), *The very model of a modern linguist — in honor of Helge Dyvik*, 260–281. Bergen: University of Bergen. 10.15845/bells.v8i1.1335.